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MILITARY STANDARD

ALUMINUM AND MAGNESIUM PRODUCTS PREPARATION FOR SHIPMENT AND STORAGE

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DEPARTMENT OF DEFENSE
Washington, D. C. 20360

Aluminum and Magnesium Products, Preparation For Shipment and Storage

MIL-STD-649C

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Recommended corrections, additions, or deletions should be addressed to Warner Robins ALC (MMED), Robins AFB, Ga. 31098.

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MILITARY STANDARD

ALUMINUM AND MAGNESIUM PRODUCTS
PREPARATION FOR SHIPMENT AND STORAGE

1. INTRODUCTION AND SCOPE

1.1 Scope. This standard describes the minimum requirements for materials, methods, containers and procedures for the preparation of aluminum and magnesium products for shipment and storage. Materials, methods and containers not listed in this standard will not be used without prior approval of the procuring or requiring activity concerned.

1.2 Application. This standard is intended for use by commercial suppliers on new procurements and by military activities which physically process aluminum and magnesium products for shipment and storage.

2. REFERENCED DOCUMENTS

2.1 Government publications. The following documents, of the issue in effect on the date of the invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

SPECIFICATIONS:

Federal

FF-N-105	Nail, Brads, Staples and Spikes, Wire, Cut and Wrought
NN-P-530	Plywood, Flat Panel
QQ-S-781	Strapping, Steel, and Seals
UU-P-553	Paper, Wrapping, Tissue
VV-L-800	Lubricating Oil, General Purpose, Preservative (Water-displacing, Low Temperature)
LLL-B-810	Building Board, (Hard Board) Hard Pressed, Vegetable Fiber
MMM-A-125	Adhesive, Casein-type, Water and Mold Resistant
PPP-B-566	Box, Folding, Paperboard
PPP-B-636	Box, Shipping, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated Triplewall
PPP-C-96	Can, Metal, 28 Gage and Lighter
PPP-D-705	Drum, Shipping and Storage: Steel, 16 and 30 Gallon Capacity
PPP-D-723	Drum, Fiber
PPP-D-729	Drum, Shipping and Storage, Steel, 55 Gallon
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-P-704	Pails, Metal: (Shipping, Steel, 1 through 12 Gallons)
PPP-S-760	Strapping, Nonmetallic, (and Connectors)
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing
PPP-T-60	Tape, Packaging, Waterproof

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PPP-T-76	Tape, Pressure-sensitive Adhesive Paper (For Carton Sealing)
PPP-T-495	Tubes, Mailing, and Filing
PPP-V-205	Veneer, Paper Overlaid, Container Grade

Military

MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-B-131	Barrier Materials, Watervaporproof, Flexible, Heat-Sealable
MIL-M-3171	Magnesium Alloy, Processes for Pretreatment and Prevention of Corrosion on
MIL-L-7870	Lubricating Oil, General Purpose, Low Temperature
MIL-C-11796	Corrosion Preventive Compound, Petrolatum, Hot Application
MIL-C-16173	Corrosion Preventive Compound, Solvent Cutback, Cold Application
MIL-P-17667	Paper, Wrapping, Chemically Neutral (Non-Corrosive)

STANDARDS:

Federal

Fed. Test Method Std. No. 101	Preservation, Packaging, and Packing Materials: Test Procedures

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40 Inch X 48 Inch Pallets, Skids, Runners or Pallet Type Base
MIL-STD-731	Quality of Wood Members For Containers and Pallets

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this standard. Unless otherwise indicated, the issue in effect on the date of the invitation for bids shall apply.

Non-governmental

PS 1-74	U. S. Product Standard (For Construction and Industrial Plywood)

(The above publication is available from National Bureau of Standards, Product Standards Section, Wash D.C. 20234)

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Official Classification Committee,
Uniform Freight Classification Rules

(The above publication is available from the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N.Y. 10016)

American Trucking Association,
National Motor Freight Classification Rules

(The above publication is available from the American Trucking Association, 1616 P Street N. W., Wash., D. C. 20036)

The Aluminum Association Standard,
Protective Oil for Aluminum

(The above publication is available from The Aluminum Association, 420 Lexington Avenue, New York, N.Y. 10017)

3. DEFINITIONS

3.1 Levels of protection. The following levels of protection apply equally to preservation and packing.

3.1.1 Level A. The degree required for protection against the most severe conditions known or anticipated to be encountered during shipment, handling and storage. These conditions include, but are not limited to multiple rough handling during transportation, intransit storage, and worldwide redistribution by DOD storage and redistribution depots from manufacturer to ultimate user; environmental exposure during transit; and extended unimproved open storage in all climatic zones.

3.1.2 Level B. The degree required for protection of material under known favorable conditions during shipment, handling and storage.

3.1.3 Level C (Commercial packaging). The degree required for protection of material during shipment from supply source to the first receiving activity for immediate use. The methods and materials employed by the supplier to satisfy the requirements of the commercial distribution system to provide protection against corrosion, deterioration and damage during shipment to a using customer may be used if it meets this requirement.

3.2 Packaging terminology.

3.2.1 Exterior pack. A container, bundle, or assembly which is sufficient by reason of material, design and construction to protect material during shipment and storage. This can be the unit pack or a container with any combination of unit or intermediate packs.

3.2.2 Intermediate pack. A wrap, box or bundle which contains two or more unit packs of identical items.

3.2.3 Marking. Application of numbers, letters, labels, tags, symbols or colors for handling or identification during shipment and storage.

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3.2.4 Packaging. The processes and procedures used to protect material from deterioration and damage. It includes cleaning, drying, preserving, packing, marking and unitization.

3.2.5 Packing. Assembling of items into a unit, intermediate or exterior pack with necessary blocking, bracing, cushioning, weatherproofing, reinforcement and marking.

3.2.6 Preservation. Application of protective measures, including cleaning, drying, preservative materials, barrier materials, cushioning and containers when necessary.

3.2.7 Unit pack. The first tie, wrap or container applied to a single item or a quantity thereof, or to a group of items of a single stock number, preserved or unpreserved, which constitutes a complete or identifiable package.

3.3 Definition of terms. The following additional definitions are related to the preparation for delivery and storage of aluminum and magnesium products.

3.3.1 Box. Rigid container having closed faces.

3.3.2 Bundle. Two or more articles fastened together to form a shipping unit.

3.3.3 Case liner. Same as wrapped, see 3.3.26.

3.3.4 Cleat. Reinforcement member fastened to the top, bottom or sides and ends of a container to insure proper assembly and added strength.

3.3.5 Container. Receptacle capable of closure.

3.3.6 Corner protector. Protective material placed under ties to protect edges of a package.

3.3.7 Crate, open. Rigid container of structural members fastened together.

3.3.8 Deckboard. Piece of lumber at right angles to stringers or skids of a pallet to form a bearing surface.

3.3.9 Edge protector. Protective material placed between sheared edges of coils of foil and flanges.

3.3.10 Filler. Piece of material placed in a package to fill void space for the purpose of squaring out the contents.

3.3.11 Framing member. Parts forming the main structure of a crate.

3.3.12 Gross weight. Weight of contents of a package including the weight of all packaging and packing materials.

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3.3.13 Header. Member of skid type base used to join the ends of 2 or more skids and provide added strength to base.

3.3.14 Interleaving. Placement of a sheet of protective material between two adjacent pieces of metal.

3.3.15 Loose. Single piece that can be handled individually.

3.3.16 Net weight. Weight of contents of a package excluding the weight of all packaging and packing material.

3.3.17 Nominal. Referring to lumber size, nominal, stated as the rough cut size, also includes sizes resulting from surfacing of the rough cut size.

3.3.18 Pallet. Low portable platform to allow the use of fork trucks or slings to facilitate handling, storage and transportation of materials as a unit. Pallets may be single or double faced.

3.3.19 Port mark. Words, symbols or colors indicating the port of destination.

3.3.20 Rubbing strips. Pieces of lumber secured to the skid portion of container or container base for the purpose of accomodating mechanical handling such as forklift entry, slings, etc.

3.3.21 Separator. Any material used in any direction for dividing and maintaining clearance within a package or a pallet load.

3.3.22 Skids. A pair or series of parallel load bearing base members affixed to the underside of boxes, crates, etc., which serves to strengthen and support the load, and to provide for mechanical handling. Skids may be longitudinal or transverse.

3.3.23 Splice. To unite or join the ends of material such as lumber, plywood or paper overlaid veneer.

3.3.24 Stringer. Member placed at right angles between the two faces or decking of a pallet.

3.3.25 Tension tied. Securement applied with mechanical tools.

3.3.26 Wrapped. Unit entirely surrounded with moisture-resistant barrier material. All joints are closed with seams, overlaps, multiple lockfolds or sealed to protect the unit against contact with water and other undesirable elements. The joints of multiple layers of protective material are closed separately and independently of each other.

4. GENERAL REQUIREMENTS

4.1 Materials, methods and containers. Materials, methods and containers shall conform to the requirements of this standard. Those exceeding the requirements herein may be substituted when approved by the Administrative Contracting Officer (ACO) or authorized representative of the requiring activity, if necessary due to the non-availability of materials or equipment.

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4.1.1 Materials which are not covered by applicable specifications or which are not specifically described herein shall be of the best quality and shall be compatible with and protect the contents.

4.1.2 Packing procedures and construction details for containers and methods not covered by specification reference shall be in accordance with 5.3.

4.1.3 Splicing requirement. When container members must be spliced to obtain the required length or width, the adjacent edges of the two pieces being spliced shall be butt jointed as specified in 4.1.3.1 and 4.1.3.2 and each piece fastened to the splice board. The fastening shall conform to the requirements specified for construction of the panels being spliced. Nails must be clinched.

4.1.3.1 To splice box boards. Splice boards shall be applied to extend on each side of the joint at least two times the width of and the same thickness as the box boards.

4.1.3.2 To splice plywood and POV. Additional cleats, as required, shall be centered over the joint. The thickness of each joint cleat (splice board) shall be as specified for edge cleats and the width shall not be less than 1 1/3 times the required width of the edge cleat, but under no circumstances shall the width be less than 2 1/4 inches.

4.1.4 The inside dimensions of boxes shall correspond with the length and cross-sectional area of the contents as closely as possible.

4.2 Internal packaging materials.

4.2.1 Material compatibility. Internal packaging materials shall not adversely affect the contents because of incompatibility of hygroscopic or chemical properties.

4.2.2 Blocking and bracing. Articles not completely filling the shipping container shall be blocked, braced, fastened or otherwise secured. Articles having projecting parts that may be broken or may puncture the container shall be rigidly supported, suspended or otherwise protected. Clearance of at least one inch shall be provided between projecting parts and the adjacent inside face of the container. Blocking and bracing shall be prevented from coming in direct contact with any unprotected surface of the item by use of Type II paper or as detailed in Table IV for a specific item.

4.3 Handling.

4.3.1 General. Containers and pallets in their shipping configuration shall be provided with lifting and hoisting provisions commensurate with their weight, size and intended mode of transportation to insure safe and efficient movement.

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4.3.2 Hoisting. Convenient means shall be provided on all shipping containers (except figure 8, 9 and 10) and pallets weighing more than 200 pounds gross which will permit hoisting by attaching suitable slings at the bottom of the containers/pallets. Hoisting points shall be placed on each end of the long dimension of the container/pallet base.

4.3.3 Forklift Truck Compatibility. Unless otherwise specified herein and except figure 8, 9 and 10 boxes, containers/pallets grossing over 200 pounds must be capable of being handled from at least two sides (fourway entry is required for containers 60 inches or more in length or width) by forklift trucks. Openings shall be a minimum of 3 inches high and at least 20 inches apart inside-to-inside, symmetrically about the center of balance. Containers may have a single opening 40 inches wide or more to provide forklift access.

5. DETAILED REQUIREMENTS

5.1 Preservation. Preservation shall be Level A, B or C (commercial preservation) as specified.

5.1.1 Level A and B. Detailed requirements for preservation of aluminum and magnesium products are listed alphabetically by product in Table I. When Level A or B is specified, items will be preserved in accordance with the detailed requirements outlined herein.

5.1.1.1 Cleaning. Surfaces shall be commercially clean, being free of loose mill scale, dirt or corrosion. The presence of residual rolling oil shall not be considered foreign matter.

5.1.1.2 Preservatives. Oil used for preservation of aluminum products shall conform to the requirements of Aluminum Association Standard, Protective Oil for Aluminum or VV-L-800 or MIL-L-7870. Oil used for preservation of magnesium products shall conform to the requirements of MIL-C-16173, Grade 2 or MIL-C-11796, Class 3. Chrome pickle treatment of magnesium products shall conform to the requirements of MIL-M-3171.

5.1.1.3 Application of preservative coating. The application of corrosion preventive material demands careful attention; compounds shall not be applied to surfaces which show signs of moisture, condensation, frost, rust, loose mill scale, dirt or other contaminants. The maximum corrosion resistance is obtained only by a thorough coating of a contaminant free surface. It is essential that articles treated are not unduly handled until film is set. Application may be by dipping, brushing, rolling, spraying or flowing onto the surface.

5.1.1.4 Interleaving. Interleaving paper, when required by Table II and III below, shall be Type I paper (Table IV). The interleaving paper shall fully cover the metal; for manual application, paper shall extend 1" beyond sides and ends, for machine application, paper may be same size as metal provided metal is fully covered.

5.1.1.5 Wraps. Application of wraps shall be in accordance with Table I. Products may be machine wrapped or hand wrapped.

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TABLE I

Preservation and Packing for Levels A or B

(NOTE: For Level C/Commercial Packaging, See 5.1.2 and 5.3.2)

Product	Preservation (5.1.1)	Packing (5.3 and 5.3.1) For Barrier, See Table IV	Maximum Net Weight Per Container (5.3.1.13)
Bar, Rod & Wire (Cold-finished, drawn, extruded, rolled, and forged):			
Coiled, Bare	Al-oiled, Mg-Chrome pickled (5.1.1.1 - 5.1.1.3)	Wrapped Coils (Fig 19). Wrap with two layers of Type IIIB barrier.	120
Coiled, Covered	None required	Wrapped Coils (Fig 19). Wrap with two layers of Type IIIB barrier.	120
Spooled (5, 10, 15, 20 or 30 pounds per spool)	None required	Wooden boxes (Fig 1-4). Boxes shall be case lined with two layers of Type II barrier.	300
Straight lengths	Al-oiled, Mg-Chrome pickled (5.1.1.1-5.1.1.3)	Wooden boxes (Fig 5-7). Boxes shall be case lined with two layers of Type II barrier.	1000 1/
		or	
		Corrugated fiberboard boxes, Class weather-resistant (5.3.1.1.2.1)	300
		or	
		Fiber tubes (5.3.1.12)	200

1/ Weight may be increased to 2000 pounds for large quantities for manufacturing when specified in the contract or order.

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TABLE I Continued

Product	Preservation (5.1.1)	Packing (5.3 and 5.3.1) For Barrier, See Table IV	Maximum Net Weight Per Container (5.3.1.13)
Blooms and Billets	See ingot	---	
Bus Conductors (cold-finished, drawn, extruded and rolled)	See bar, straight lengths	---	
Cable (bare and covered):			
Size 1/0 and smaller	None required	Wrapped coils (Fig 19). Wrap with two layers of Type IIIB barrier. or Reels (Fig 24).	Bare - 250 Covered-200 Bare-1250 Covered-1000
Size larger than 1/0	None required	Reels (Fig 24).	Bare-1600 Covered-1300
Castings & forgings, finished.	None required	Wooden boxes (Fig 1-4) or Style 1 crate (Fig 20). Boxes and crates shall be case lined with two layers of Type II barrier.	2000
Castings & forgings, rough.	None required	Bare bundles (Fig 23).	1000
Conduit	See ASA schedule pipe	---	

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TABLE I Continued

Product	Preservation (5.1.1)	Packing (5.3 and 5.3.1) For Barrier, See Table IV	Maximum Net Weight Per Container (5.3.1.13)
Extruded shapes (Metal less than 1 pound per linear foot) 2/	Al-oiled, Mg-Chrome pickled (5.1.1.1-5.1.1.3)	Wooden boxes (Fig 5-7). Boxes shall be case lined with two layers of Type II barrier.	2000
		or	
		Corrugated fiberboard boxes Class weather-resistant (5.3.1.1.2.1)	300
Fittings (Pipe and Conduit)	Al-oiled, Mg-Chrome pickled (5.1.1.1-5.1.1.3). External threads shall be covered with suitable thread protectors.	Wooden boxes (Fig 1-4) or Style 1 crate (Fig 20) dependent upon size of fittings. Boxes and crates shall be case lined with 2 layers of Type II barrier.	100
Foil: 3/ 4/			
Coiled	Foil shall be wound on aluminum fiber or steel cores. End of coil shall be secured with pressure sensitive tape. Sheared edges shall be protected from flanges and adjacent coils with suitable edge protectors. Each coil, or coils (see 5.3.1.6) shall be wrapped with minimum 0.001 inch thick aluminum foil.	Wooden boxes (Fig 1-4). Coils shall be suspended by extended cores or wood dowels through the core. Core extension or dowel shall be inserted in wood flanges so that periphery of coil does not contact inner surface of box (Fig 25). Minimum flange thickness shall be as specified in 5.3.1.6.	500
		or	

2/ For metal weighing one pound per linear foot or more, see structural shapes.
3/ Foil is sheet metal less than 0.006 inch thick.
4/ Foil for food handling applications shall be preserved and packed as specified in the contract or order.

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TABLE I Continued

Product	Preservation (5.1.1)	Packing (5.3 and 5.3.1) For Barrier, See Table IV	Maximum Net Weight Per Container (5.3.1.13)
Coiled (continued)	Foil wrap shall be a conformable wrap completely enclosing the coil or coils and edge protectors on each core or dowel (Fig 25).	Corrugated fiberboard boxes, class weather-resistant (5.3.1.1.2.1), suspended as above.	300
Flat	None required	Wooden boxes (Fig 1-4). Contents shall be wrapped with two separate layers of Type II barrier.	500
		or	
		Corrugated fiberboard boxes, class weather resistant (5.3.1.1.2.1)	300
Forgings	See Castings	- - - -	
Forging Stock	See Bar	- - - -	
Impact Extrusions	None Required	Wooden boxes (Fig 1-4). Boxes shall be lined with two layers of Type II barrier.	700
		or	
		Corrugated fiberboard boxes, class weather resistant (5.3.1.1.2.1).	300
Ingots:			
500 pounds per piece and over.	None required	Loose	---

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TABLE I Continued

Product	Preservation (5.1.1)	Packing (5.3 and 5.3.1) For Barrier, See Table IV	Maximum Net Weight Per Container (5.3.1.13)
30 pounds to 500 pounds per piece	None required	Bare bundles (Fig 27). Size of bundle straps shall be as shown in Table IX. A minimum of two straps shall be used per bundle.	3500
Less than 30 pounds per piece	None required	Pallets (Fig 26). Size of pallet straps shall be as shown in Table IX	2500
		or	
		Self-palletized bundle. Interlocking ingots that are self-palletizing may be shipped in strapped bundles not over 42 inches high. Bundle shall be strapped with a minimum of one 3/4 inch steel strap.	1500
Grained and granulated ingot and shot	Product to be packed in wood boxes (Fig 1-4), shall be packaged in PPP-B-566, Boxes, Folding, Paper-board.	Wooden Boxes (Fig 1-4).	500
		or	
		PPP-P-704 Steel Pails (5.3.1.8). Pails shall not be overpacked.	70
		or	
		PPP-D-705 and PPP-D-729 Steel Drums (5.3.1.9). Drums shall not be overpacked.	650
		or	
		PPP-D-723 Fiber Drums (5.3.1.11). Drums shall not be overpacked.	550

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TABLE I Continued

Product	Preservation (5.1.1)	Packing (5.3 and 5.3.1) For Barrier, See Table IV	Maximum Net Weight Per Container (5.3.1.13)
Paste and Powder	Product to be packed in wooden boxes (Fig 1-4) shall be packaged in I, 2 or 10 pound friction top can per PPP-C-96, Type V, Class 2.	Wooden boxes (Fig 1-4). or PPP-D-705 or PPP-D-729 Steel Drums (5.3.1.9). Drums shall not be overpacked.	50 600
Plate: 5/			
Flat and Tapered	Al-see Table II, Mg-see Table III.	Pallets (Fig 15-18) with pallet enclosures (Fig 12-14). Contents shall be wrapped with one layer of Type IIA and two layers of Type IIIA barrier.	4000
Circles	Al-see Table II, Mg-see Table III.	Pallets (Fig 15-18) with pallet enclosures (Fig 14). Contents shall be wrapped with two layers of Type II barrier.	4000
Floor and Tread Plate and Abrasive Tread Plate	None required.	Pallets (Fig 15-18). Secure contents to pallet with minimum two lengthwise and two girthwise straps, size 1 1/4" X 0.031".	10000
Screw Machine Stock	See bar	- - -	

5/ Classification of Sheet and Plate: Sheet is 0.006 through 0.249 inch thick; Plate is 0.250 inch and thicker.

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TABLE I Continued

Product	Preservation (5.1.1)	Packing (5.3 and 5.3.1) For Barrier, See Table IV	Maximum Net Weight Per Container (5.3.1.13)
Sheet: 5/			
Flat and Tapered:			
90 pounds per piece or less or 15 feet in length or less	Al-See Table II, Mg-see Table III	Boxes, wood, plywood, POV, solid fiberboard panel (Fig 8-11). Contents of box shall be wrapped with one layer of Type IIA and one layer of Type III barrier. When quantity exceeds one box to a single destination, boxes shall be strapped to a pallet (5.3.1.1.1) not exceeding 4000 pounds net weight.	500
Over 90 pounds per piece or over 15 feet in length	Al-See Table II, Mg-see Table III	Pallets (Fig 15-18) with pallet enclosures (Fig 12-14). Contents shall be wrapped with one layer of Type IIA and two layers of Type IIIA barrier.	4000
Coiled	Al-oiled, Mg-Chrome pickled (5.1.1.1-5.1.1.3)	Pallets (Fig 15-18) with pallet enclosure (Fig 14). Contents shall be wrapped with two layers of Type II barrier.	4000
Roofing and siding	None required	Pallets (Fig 15-18) with pallet enclosures (Fig 12-14). Contents shall be wrapped with one layer of Type IIA and two layers of Type IIIA barrier.	4000

5/ Classification of sheet and plate: Sheet is 0.006 through 0.249 inch thick; plate is 0.250 inch and thicker.

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TABLE I Continued

Product	Preservation (5.1.1)	Packing (5.3 and 5.3.1) For Barrier, See Table IV	Maximum Net Weight Per Container (5.3.1.13)
Structural Shapes (Extruded & rolled). 6/			
Less than 150 pounds per piece	Al-none required, Mg-Chrome pickled.	Wooden Boxes (Fig 5-7). Boxes shall be case lined with two layers of Type II barrier.	1000
		or	
		Corrugated Fiberboard boxes, Class weather-resistant (5.3.1.1.2.1).	300
150 pounds per piece and over	Al-none required, Mg-Chrome pickled.	Loose	---
Tubular Products (Drawn, extruded and welded):			
Coiled	Al-oiled, Mg-oiled (5.1.1.1-5.1.1.3). Stagger wound coils shall be tied through the core in two places with twine or pressure sensitive tape. Pan-cake or level layer wound coils shall not be tied.	Style 1 crates (Fig 20). Crates shall be case lined with one layer of Type II and one layer of Type III barrier.	700

6/ For metal weighing less than one pound per linear foot, see extruded shapes.

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TABLE I Continued

Product	Preservation (5.1.1)	Packing (5.3 and 5.3.1) For Barrier, See Table IV	Maximum Net Weight Per Container (5.3.1.13)
Straight lengths	Al-oiled, Mg-Chrome pickled (5.1.1.1-5.1.1.3). When the wall thickness of tube is less than 2 1/2% of the outside diameter, boxes shall be lined with suitable cushioning material.	Wooden Boxes (Fig 5-7). Boxes shall be case lined with two layers of Type II barrier.	1000
		or	
		Corrugated fiberboard boxes class weather resistant (5.3.1.1.2.1)	300
		or	
		Fiber Tubes (5.3.1.12).	200
ASA Schedule Pipe	Al-no preservative required, Mg-Chrome pickled (5.1.1.1-5.1.1.3). External threads shall be covered with suitable thread protectors.	Wooden Boxes (Fig 5-7). Boxes shall be case lined with two layers of Type II barrier.	800
		or	
		Style 2 and 3 crates (Fig 21-22). Crates shall be case lined with one layer of Type IIIA barrier.	4000
		or	
		Fiber Tubes (5.3.1.12).	200
Construction Pipe	Al-no preservative required, Mg-Chrome pickled (5.1.1.1-5.1.1.3).	Same as for ASA Scheduled Pipe above.	See Packing
Welding and Brazing Rod:			
Coiled	None required	Wrapped coils (Fig 19). Wrap with two layers of Type II barrier.	120

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TABLE I Continued

Product	Preservation (5.1.1)	Packing (5.3 and 5.3.1) For Barrier, See Table IV	Maximum Net Weight Per Container (5.3.1.13)
Straight lengths, 36 inch	Package 5 pounds per fiber tube with metal ends or 10 pounds per fiberboard carton. 10 fiber tubes or 8 cartons shall be overpacked in a weather-resistant fiberboard carton IAW PPP-B-636. All corners and seams of boxes, including manufacturer's joint, shall be sealed with PPP-T-60, Type III or IV, Class 1, minimum 2 inch wide tape.	Wooden Boxes (Fig 1-4). Fiberboard boxes shall be overpacked in wooden boxes.	1000
Inert gas welding electrode	1 pound, 5 pound, 10 pound, 12 1/2-15 pound or 30 pound spools. Individual spools shall be adequately protected by application of moisture-resistant barrier and packaged single or in multiples in fiberboard boxes IAW PPP-B-636, Type CF, Class weather-resistant.	Fiberboard boxes per PPP-B-636, Type CF, Class weather resistant, Grade V3c. All corners and seams of boxes, including manufacturer's joint, shall be sealed with PPP-T-60, Type III or IV, Class 1, minimum 2 inch wide tape. Three 5/8 inch X 0.015 inch flat steel straps, 1 lengthwise and 2 girthwise, shall be applied to each box. Maximum gross weight and dimensions of the box specification shall not be exceeded.	See Packing
		or	
		Wooden Boxes (Fig 1-4). Intermediate fiberboard boxes overpacked in wooden boxes, shall be closed IAW the closure method prescribed above for fiberboard shipping containers.	1000

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TABLE II

Preservation of Aluminum Sheet and Plate (See Table I)

Product	Preservation 1/
Abrasive tread plate	None required
Flat and tapered sheet & plate, and all circles:	
Clad, all alloys:	
Circles less than 12" diameter	None required
Flat, tapered & circles 12" diameter and over	Interleaved
Nonclad:	
Circles less than 12" diameter	None required
Flat, tapered & circles 12" diameter and over:	
Heat-treatable alloys, all tempers 2/	Oiled (5.1.1.1-5.1.1.4)
Non Heat-treatable alloys: 2/	
Annealed	Interleaved
Other than annealed: 0.060" thick and over	Interleaved
Less than 0.060" thick	Oiled (5.1.1.1-5.1.1.4)
Painted, chemical conversion coated and anodized sheet and plate	Interleaved
Floor and tread plate	None required
Coiled sheet	Oiled (5.1.1.1-5.1.1.4)
Perforated sheet	Oiled (5.1.1.1-5.1.1.4)
Roofing and siding sheet	None required

1/ Interleaving paper shall be Type 1 paper (Table IV). For metal 0.090 inches thick and less interleaving paper shall be minimum 10 pound basis weight; for metal over 0.090 inches thick interleaving paper shall be minimum 15 pound basis weight. The interleaving paper shall fully cover the metal; for manual application paper shall extend 1" beyond sides and ends, for machine application paper may be same size as metal provided metal is fully covered.

2/ Heat treatable alloys are the 2000, 6000 and 7000 groups. Non heat-treatable alloys are the 1000, 3000 and 5000 groups.

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TABLE III

Preservation of Magnesium Sheet and Plate (See Table I)

Product	Preservation 1/ 2/
Flat sheet and plate:	
Less than 0.005 inch thick	Oiled (5.1.1.1-5.1.1.4) or Oiled and interleaved (5.1.1.1-5.1.1.4) or Chrome pickled and interleaved
0.005 inch and thicker	None or Oiled (5.1.1.1-5.1.1.4)

1/ Interleaving paper shall be as specified for aluminum sheet and plate (Note 1, Table II)
2/ Preservation desired must be as specified in the contract or order.

5.1.2 Level C (commercial preservation). When Level C or Commercial preservation is specified, items will be given the best degree of protection employed by the supplier to afford protection against corrosion, deterioration and damage during shipment.

5.2 Material requirements.

5.2.1 Fiberboard. Fiberboard for boxes and liners shall conform to the requirements of PPP-F-320, class weather-resistant.

5.2.2 Paper and barrier material. Paper and barrier material shall conform to the requirements of Table IV below.

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TABLE IV

Minimum Requirements for Paper and Barrier Materials (See Table I) 1/

Type	Tensile Strength in Weaker Direction Meth 2038 Pounds	Stretch in Creped Direction Meth 2038 Percent	Contact Corrosivity Meth 3005	Oil Resistance Meth 3017	Water Resistance Dry Indicator Method Hours	Military/ Federal Specification
I. Interleaving 2/						
A. 10 lbs, uncreped	1 1/2		No	No delamination, embrittlement or disintegration		UU-P-553, Tp II
B. 15 lbs, uncreped	5		Corrosion			MIL-P-17667, Tp I
II. Exterior & Interior Wrap Non Reinforced 3/				No Penetration for Min 24 hrs & No delamination, embrittlement or disintegration		MIL-B-121
A. Uncreped	30	-	No		24	Tp 1, Gd A, Cl 1
B. Creped (Stretchable)	25	15	Corrosion		24	Tp 1, Gd A, Cl 2
III. Exterior & Interior Wrap, Reinforced 3/						
A. Uncreped 5/	65		No		24	
B. Creped 4/ 6/	35	20	Corrosion	"	24	

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TABLE IV Continued

NOTES:

1/ The minimum requirements listed in Table IV are based on the results of tests performed in accordance with the methods outlined in FED-STD-101, except water resistance, or shall meet the minimum requirements of the applicable Military/Federal specification cited. Water resistance test is based on test method D779 of American Society of Testing and Materials (ASTM) Standards, Part 15.

2/ Interleaving paper shall be non-abrasive to aluminum surfaces, and have a Hydrogen Ion concentration (pH) of between 4.5 and 7.5.

3/ Asphalt paper shall not be used.

4/ Shall have random dispersed reinforcement, or a reinforcement spacing not less than 12 threads per foot in the longitudinal direction.

5/ Shall have random dispersed reinforced, or a reinforcement spacing not less than 12 threads per foot in both directions.

6/ Tensile strength shall apply only in the reinforced direction.

5.2.3 Wood. The species of wood indicated in Table V and classified into groups as shown shall be used. The groups are set up to include, in any one group, species of woods that have approximately similar characteristics important to box design. These characteristics include density, flexural and compressive strength, stiffness, shock absorption and nail holding power. When any species of wood is specified, any one or more species in the same group may be used. Groups I and II may be interchanged. Groups III and IV may be interchanged.

TABLE V

Species of Wood (Ref MIL-STD-731)

Group I	Group II	Group III	Group IV
Alder, red	Douglas Fir	Ash (Except White Ash)	Ash (White)
Aspen (Popple)	Hemlock	Elm Soft	Beech
Basswood	Larch (Tamarack)	Gum, Red or Black	Birch
Buckeye	Pine, So. Yellow	Maple, Soft or Calif.	Cherry
Butternut	Western Larch	Oak, Calif.	Elm, Hard
Cedar		Sweetgum	Hackberry
Chestnut		Sycamore	Hickory
Cottonwood		Tupelo	Locust
Cypress			Maple, Hard
Fir (Abies sp)			Oak
Magnolia			Pecan
Pine (except So. Yellow)			
Redwood			
Spruce			
Willow			
Yellow Poplar			

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5.2.4 Lumber. All lumber dimensions used in this standard are nominal sizes except where minimum sizes are specified. At least one surface, which is to be placed on the outside of boxes, crates, or pallet closures, shall be sufficiently smooth to permit legible marking, stencilling or printing. Pieces shall be cut true to length. The boards shall be free from defects that materially weaken them, expose the contents of the box to damage or interfere with the prescribed fabrication or nailing.

5.2.5 Plywood. Unless otherwise specified, plywood used to fabricate cleated panels shall conform to PS 1-74 and NN-P-530, standard interior (Grade C-D) with exterior glue.

5.2.6 Paper overlaid veneer (POV). Unless otherwise specified, POV used for boxes shall be in accordance with PPP-V-205, Type II.

5.2.7 Hardboard. Unless otherwise specified, hardboard shall be in accordance with LLL-B-810, Type I, Surface I, Finish A, Design A.

5.2.8 Nails and staples. Nails and staples shall conform to FF-N-105. Except for Style 18, unclinched nails shall be cement coated or chemically etched.

5.2.9 Strapping. Flat and round steel strapping shall conform to QQ-S-781, Class 1, Finish A (coated finish) and Class 2, Type V, Finish B (zinc-coated) respectively. Equivalent non-metallic strapping conforming to PPP-S-760 may be used when comparable strength requirements are satisfied.

5.2.10 Tapes. Tapes shall conform to the requirements of PPP-T-45, PPP-T-60 or PPP-T-76 as applicable.

5.3 Packing. Only one type, class or size of material or National Stock Number (NSN) shall be packed in one container. Under no circumstances will more than one NSN be packed within the unit or shipping container. This requirement does not prevent consolidation of two or more shipping containers on pallets to facilitate handling. Packing shall be Level A, B or C (commercial packing) as specified.

5.3.1 Level A and B. Detailed requirements for packing aluminum and magnesium products are listed alphabetically by product in Table I. When Table I provides a choice of several containers for a particular product, any one of the containers may be selected for use unless specifically prohibited by the contract or order except, for the Army only, when Table I provides a choice between wood and fiberboard boxes and Level A applies, only wood boxes will be used. This section provides requirements applicable to packing procedures and construction details for containers and methods not covered by specification reference. When Level A or B is specified, items will be packed in accordance with the detailed requirements outlined herein.

5.3.1.1 Boxes (Figure 1 through 11).

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5.3.1.1.1 Boxes for flat sheet. (Figures 8, 9, 10 and 11). When required by Table 1, material shall be boxed in wood (Figure 8), plywood or paper overlaid veneer (Figure 9) or solid fiberboard panel (Figure 10 and 11) boxes. Quantity per box shall be limited to 500 pounds net weight. When shipment exceeds one box to a single destination, boxes shall be strapped to pallets (5.3.1.2.1) conforming to Figures 15, 16, 17 or 18 as applicable with a minimum of three girthwise 1 1/4" X .031" steel straps. Not less than two lengthwise 1 1/4" X .031" steel straps shall be tension-tied around the total load being palletized. Straps shall pass over existing top cleats or 1" X 2" battens shall be applied in the void and secured to the strapping with staples.

5.3.1.1.1.1 Top and bottom panels. Top and bottom panels shall consist of a complete covering of lumber, plywood, paper overlaid veneer or a single piece of V2s solid sheet fiberboard.

5.3.1.1.1.2 Side and end panels. The thickness of the side and end panels for Figure 8-11 boxes shall be such that the inside depth of the box corresponds with the thickness of the contents without voids.

5.3.1.1.1.2.1 Figure 8 box. Side and end panels for Figure 8 box shall consist of lumber attached to the top and bottom container panels to effect a telescopic closure. Lumber attached to the underside of the top panel shall be fastened flush with all outer edges. Lumber attached to the bottom panel shall be set in from the outer edges a distance sufficient to result in approximately 1/4 inch telescopic clearance to facilitate easy container closure. Depth thickness of side and end members shall be varied to obtain the desired depth of box not to exceed the maximum depth of Table VI. The total width of the side and end panels, including maximum 1/4 inch telescoping tolerance, shall not be less than the width established in Table VI when the container is closed. Side and end panels shall be nailed or stapled.

5.3.1.1.1.2.2 Figure 9 box. Side and end panels for Figure 9 box shall consist of lumber attached to the bottom container panel edge cleats as shown in detail A and B. The side and end panels shall be placed approximately 1/4 inch from the bottom panel to prevent binding when the top is placed on the container. Thickness of the side and end members shall be varied to obtain the desired depth of box not to exceed the maximum depth of Table VI. The width of the side and end panels, including telescoping tolerance, shall be not less than the width established in Table VI when the container is closed. Side and end panels shall be nailed or stapled.

5.3.1.1.1.2.3 Figure 10 box. Side and end panels for Figure 10 box shall be constructed as specified in 5.3.1.1.1.2.1 except that sides and ends shall be stapled.

5.3.1.1.1.2.4 Figure 11 box. Side and end panels for Figure 11 box shall consist of lumber attached to the top and bottom container panels to effect a telescopic closure. Side and end panels attached to the underside of the top panel shall be fabricated from maximum 3/4 X 3/4 inch lumber and shall be fastened flush with all outer edges. The bottom side and end panels shall be of 2 piece assembly, constructed by lamination 3/4 X 3/4 inch maximum dimension lumber to a lumber spacer not more than 3/4 X 1 1/2 inches in size as shown in detail A and B. The laminated assembly shall be attached to the bottom container panel, flush with all outer edges. Side and end panels shall be stapled.

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5.3.1.1.1.3 Cleats. Top and bottom panels shall be provided with cleats in accordance with Table VI. Cleats, except for Figure 10 and 11, shall be nailed or stapled. Figure 10 and 11 edge cleats shall be stapled and glue laminated, intermediate cleats shall be glued only. Any water-resistant adhesive which provides a bond equivalent in strength to the material being bonded may be used.

5.3.1.1.1.3.1 Positioning of cleats. Lengthwise cleats shall extend the full length of the panel except Figure 8 box will require no lengthwise cleats. Figure 9 lengthwise and girthwise edge cleats shall be offset as shown in the detail a distance the width of the side and end panel plus 1/4 inch. Figure 8 girthwise edge cleats and Figure 10 and 11 lengthwise and girthwise edge cleats shall be flush with the outer edges of the top and bottom panels. Intermediate girthwise cleats, of the same width and thickness as the edge cleats, shall be placed a maximum of every 48 inches center-to-center. When top and bottom panels, Figure 9 container, require girthwise splicing to facilitate covering of metal sheets more than 48 inches wide, the widths of the outer lengthwise edge cleats may exceed the requirements of Table VI and incorporate the splicing requirements of 4.1.3.

5.3.1.1.1.4 Nailing and stapling. Size and spacing of nails and staples shall be in accordance with Tables VII and VIII. Nails or staples used to attach Figure 8 and 9 intermediate cleats shall be slightly overdriven in the sequence of through the panel and into the cleat and shall be clinched at least 1/8 inch. Intermediate cleats for Figure 10 and 11 box shall be fastened in accordance with 5.3.1.1.1.3. The sides and ends shall be stapled or nailed as required (see 5.3.1.1.1.3) to the top and bottom container panels with the same staples or nails used to attach cleats to the outer container edges of the top and bottom panels specified in 5.3.1.1.1.2 and 5.3.1.1.1.3. Staples and nails shall be driven in the sequence of through the cleats and into the sides and ends. No staple or nail shall be driven completely through the sides and ends. As an alternate for Figure 8 and 9 boxes only, staples and nails may be driven in the sequence of through the sides and ends and into the cleats and shall be clinched at least 1/8 inch. The positioning of the staples and nails shall not interfere with the container closure or come in contact with the packaged contents.

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TABLE VI

Boxes for Flat Sheet (5.3.1.1.1)

Style of Box (Figure Number)	Depth of Box (Inside) Inches	Sides & Ends (Min Thickness) Inches 1/	Top & Bottom Thickness Inches	Cleats (Min Thick & Width) Inches
8	2 1/2	1 1/2	1/2 ± 1/16	1/2 X 2 1/2
9	2 1/2	1	3/16 ± 1/16	1/2 X 2 1/2
10	See Note 2	1 1/2	.090	5/16 X 2 1/2
11	1 1/2 (includes thickness of spacer)	1 1/2	.090	5/16 X 2 1/2

1/ Thickness is that dimension measured from inside to outside of box. Includes telescoping tolerances.
2/ Boxes not exceeding 8 feet in length, 1 1/2 inch; boxes exceeding 8 feet in length, 3/4 inch.

TABLE VII

Average Nail and Staple Spacing for Boxes for Flat Sheet

PARTS FASTENED	SPACING (INCHES) 1/
Top and bottom to intermediate cleats on wood boxes (Figure 8)	4
Top and bottom to intermediate cleats on plywood and POV boxes (Figure 9)	12
Girthwise edge cleats and top and bottom to side and end panels on wood boxes (Figure 8)	9
Edge cleats to top and bottom and side and end panels on plywood and POV boxes (Figure 9)	9
Edge cleats to top and bottom and side and end panels on solid fiberboard panel boxes (Figure 10 & 11)	9

1/ Not less than one nail or staple shall be used at end of all cleats except Figure 10 & 11 intermediate cleats.

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TABLE VIII

Nail and Staple Sizes for Boxes for Flat Sheet

PARTS NAILED OR STAPLED	NAIL OR STAPLE SIZE
Top and bottom to intermediate cleats (Figure 8 and 9)	2 d 1/
Edge cleats and top and bottom to side and end panels:	
(Figure 8 and 9)	5 d to 12 d 2/
(Figure 10 & 11)	0.0625 (16 gage) 3/

1/ 2 d nails or minimum 0.0625 inch diameter (16 gage) staples having minimum 7/16 inch crown with leg lengths equal to the sum of the thickness of the panels and cleats plus 1/8 inch and clinched.
2/ Size of nail varies with the thickness of the top and bottom and side or end panel.
3/ Length of staple shall be equal to thickness of cleat, panels and 1/2 to 7/8th of the height of the side and end.

5.3.1.1.1.5 Placing material in container. When placing flat sheet in container, care shall be exercised to insure that the surface containing item identification markings is placed upright for easy identification when top is removed.

5.3.1.1.1.6 Closure (Strapping). Figure 8, 9, 10 and 11 boxes shall be closed by strapping positioned at all girthwise cleats. Size of straps shall be in accordance with Table IX.

TABLE IX

Strap Sizes for Boxes and Bundles (5.2.9)
Including Ingot Bundles (Figures 1-11, 19, 24, 26 and 27)

Type of Package or (Ingots)	Weight of Contents (Pounds)	Flat Steel Strapping (Inch)	Round Steel Strapping (Inch)
Pallets	--	5/8 X 0.020	0.0800 (14 gage)
--	Less than 700	5/8 X 0.020	0.0800 (14 gage)
Bundles	--	3/4 X 0.023	0.1055 (12 gage)
--	700 to 2,000	3/4 X 0.023	0.1055 (12 gage)

5.3.1.1.2 Boxes for products other than flat sheet. When required by Table I, products shall be packed in wood boxes (Figure 1-7) and corrugated weather-resistant fiberboard boxes.

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5.3.1.1.2.1 Construction of corrugated weather-resistant fiberboard boxes. Triplewall corrugated weather-resistant fiberboard boxes shall be constructed in accordance with PPP-B-640. Singlewall and doublewall corrugated weather-resistant fiberboard boxes shall be constructed in accordance with PPP-B-636. Maximum inside dimensions of box (length, width, and depth added) shall not exceed 215 inches. Weight limitations of Table I shall apply.

5.3.1.1.2.2 Construction of wood boxes (Figure 1-7).

5.3.1.1.2.2.1 Top and bottom panels. Top and bottom panels shall consist of a complete covering of lumber. Panels for boxes, Figure 1 through 5 shall be without cleats. Panels for boxes, Figure 6 and 7 shall be with cleats. Thickness of panels shall be in accordance with Table X.

5.3.1.1.2.2.2 Side and end panels. Side and end panels shall consist of a complete covering of lumber, unless otherwise specified. Thickness of panels shall be in accordance with Table X.

5.3.1.1.2.2.2.1 Panels for boxes, Figures 1 and 5, 11 1/2 inches or less in depth and not more than 16 feet long shall be of one piece construction and without cleats, except that end panels may be two piece, cross grain laminated to obtain the required thickness. Side panels more than 16 feet long may be of two piece butt joint and splice construction (4.1.3) without cleats.

5.3.1.1.2.2.2.2 Panels for boxes, Figures 2, 3, 4, 6 and 7, more than 11 1/2 inches in depth shall be with cleats, except for side panels for boxes Figure 2, 3 and 4.

5.3.1.1.2.2.3 Cleats. When required by 5.3.1.1.2.2.1 and 5.3.1.1.2.2.2, cleats shall be in accordance with Table X. Cleats shall be nailed or stapled.

5.3.1.1.2.2.3.1 Positioning of cleats. Cleats for end panels, Figure 2 and 4, shall be placed across the grain of the panel board coverings and extend to within one-eighth inch of the inside surface of the top and bottom panels when the containers are closed. Cleats for end panels, Figure 3, shall be placed across the grain of the panel board coverings and extend to within one-eighth inch of the outside surface of the top and bottom panels when the containers are closed. Cleats for Figures 6 and 7 containers shall be set in not more than 6 inches from the container ends and spaced a maximum of 65 inches apart.

5.3.1.1.2.2.4 Box skids. Boxes with a gross weight of 200 pounds or over shall be provided with a minimum of two nominal 4 X 4 inch skids. As an alternate, skids may be fabricated from two pieces of laminated 2 X 4 inch (nominal) lumber. Skids shall replace exterior cleats on box bottom when cleats are required. Skids shall be placed parallel to and extend the full width of the box and shall be positioned not closer than 2 1/2 inches nor more than one sixth the length of the box from each end of the box. The distance between skids measured between the inside edges shall not exceed 65 inches. Additional skids, as required, shall be positioned so as to divide the distance between the end skids into units of equal length. The skids shall be notched, as applicable, to provide clearance for strapping. The skids shall be secured to the box by nails specified in 5.2.8. The nails

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shall be driven from the inside through the bottom into the skids (in two staggered rows) and shall penetrate a minimum of 3/4 the thickness of the skids. As an alternate to size specified in Table XII, nails conforming to type II, style 18 of FF-N-105 may be used of adequate length to penetrate a minimum of 3/4 the thickness of the skids and shall not protrude through the skids. The bottom half of skid ends shall be beveled at an angle of 45° plus or minus 5°.

5.3.1.1.2.2.5 Nailing and stapling. Size and spacing of nails and staples shall be in accordance with Tables XI and XII. When applying cleats, nails shall be driven through the lumber and into the cleats and clinched at least 1/8 inch.

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TABLE X

Wooden Boxes for Products Other Than Flat Sheet 1/ (Groups I and II Woods) 2/ (Figs 1-7)

Weight of Contents	Length of Box	Style of Box	Figure Number	Sides, Top & bottom (minimum thickness)	Ends (min. thick)	End cleats (minimum thickness & width)	Additional cleats (min. thickness & width)
Pounds	Inches			Inch	Inches	Inches	Inches
Less than 350	Less than 60	1	1	1/2	1 1/2	-	-
Less than 350	60 & over	1	5	5/8	1 1/2	-	-
Less than 350	Less than 60	2, 4 or 5	2, 3 & 4	1/2	3/4	3/4 x 2 1/2	-
Less than 350	60 and over	2 or 4	6 & 7	5/8	3/4	3/4 x 2 1/2	3/4 x 2 1/2
350 to 700	Less than 60	1	1	1/2	1 1/2	-	-
350 to 700	60 and over	1	5	5/8	1 1/2	-	-
350 to 700	Less than 60	2, 4 or 5	2, 3 & 4	5/8	3/4	3/4 x 2 1/2	-
350 to 700	60 and over	2 or 4	6 & 7	5/8	3/4	3/4 x 2 1/2	3/4 x 2 1/2
700 to 2,000	Less than 60	1	1	5/8	1 1/2	-	-
700 to 2,000	60 and over	1	5	11/16	1 1/2	-	-
700 to 2,000	Less than 60	2, 4 or 5	2, 3 & 4	5/8	3/4	3/4 x 2 1/2	-
700 to 2,000	60 and over	2 or 4	6 & 7	11/16	3/4	3/4 x 2 1/2	3/4 x 2 1/2

1/ Includes circles less than 36 inches diameter. For circles 36 inches diameter and over see Table VI.
2/ When groups III and IV are used in lieu of groups I and II, lumber may be one-sixteenth inch less in thickness than that specified for groups I and II.

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TABLE XI

Average Nail Spacing for Boxes of Products other than Flat Sheet and Pallet Enclosures 1/ (See Figs 1-7 & 14)

Parts Nailed	Nail Spacing Inches
Sides, top and bottom to additional cleats	4
End cleats to ends	2
Sides, top and bottom to ends and end cleats	2
Top and bottom to sides	10
Bottom to skids	3

1/ Includes circles less than 36 inches diameter. For circles 36 inches diameter and over see Table VII.

TABLE XII

Nail Sizes for Boxes of Products other than Flat Sheet and Pallet Enclosures 1/ (See Figs 1-7 & 14)

Parts Nailed	Thickness of Parts Holding Nail Heads Inch	Min Nail Size
Sides, top and bottom to additional cleats	1/2	4d
	5/8	5d
	11/16	5d
End cleats to ends	3/4	5d
Sides, top and bottom to ends and end cleats	1/2	6d
	5/8	7d
	11/16	8d
Top and bottom to sides 2/	11/16	7d
Bottom to skids	11/16	12d

1/ Includes circles less than 36" diameter. For circles 36" and over see Table VIII.

2/ Top and bottom are not nailed to sides less than 11/16" thick.

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5.3.1.1.2.2.6 Strapping. Size of strapping shall not be less than as shown in Table IX.

5.3.1.1.2.2.6.1 Strap spacing. Straps shall be placed over all girthwise cleats. On boxes without girthwise cleats, Figure 1 through 5, girthwise straps shall be set in from each end a distance equal to one-sixth the length of the box or 6 inches, whichever distance is smaller, and spaced at intervals not exceeding 65 inches (Figure 5).

5.3.1.1.2.2.7 Palletized shipment. Unless otherwise specified, boxes not exceeding 48 inches long by 40 inches wide may be palletized for shipment in accordance with the requirements of MIL-STD-147 in lieu of skid requirements of 5.3.1.1.2.2.4.

5.3.1.2 Pallets with pallet enclosures (Figures 12 through 18).

5.3.1.2.1 Pallet construction (Figures 13, 15, 16, 17 and 18).

5.3.1.2.1.1 Skid ends. Skid ends shall be cut with a dado or beveled unless rubbing strips are required in accordance with 5.3.1.2.1.3. When straps are placed over skids, skids shall be notched out to retain strapping.

5.3.1.2.1.2 Lengthwise skids, Figure 15. For pallets 52 inches wide or less, two skids shall be used and for pallets over 52 inches wide, three skids shall be used. When two skids are required, the skids shall be positioned so that the center of the second skid shall be no more than 38 inches from the outside edge. When three skids are required, one skid shall be placed in the center of the pallet. Skids shall be fabricated from nominal 4 X 4 inch, group III or IV wood.

5.3.1.2.1.2.1 Headers. Headers shall be fabricated from nominal 4 X 4 inch lumber and shall be bolted to skid ends (Figure 15) with 1/2 X 8 inch carriage bolts. Skids shall be of sufficient length for metal to be placed between the headers.

5.3.1.2.1.3 Rubbing strips. Rubbing strips shall be added to pallets (Fig 15) 60 inches or more in length. Rubbing strips shall be a minimum 3 inches high by 3 inches wide, fabricated from laminated 2 X 4 inch lumber and shall be attached to skid in such a manner as to provide 4 way entry for materials handling equipment. The bottom half of rubbing strip ends shall be beveled at an angle of 45° plus or minus 5°.

5.3.1.2.1.4 Crosswise Skids. Figure 16 illustrates the general arrangement for use on skids perpendicular to the deck boards. The skid dimensions shall be a minimum 3 inches high by 3 inches wide. Metal less than 5 feet long shall be packed on pallets using crosswise skids.

5.3.1.2.1.5 Double faced pallets (Figure 17). Skids shall be 3 X 4 inch or 2 X 6 inch lumber or may be constructed of laminated 1 X 6 inch or spliced 2 X 6 inch lumber. Joints in spliced 2 X 6 inch skids shall be reinforced with 2 X 6 inch splice boards, long enough to contact the three stringers as shown in Figure 18. Stringers shall be minimum 3 X 3 inch and may be fabricated from 4 X 4 inch or laminated 2 X 4 inch lumber. The end stringers shall be placed not more than 1 inch from the ends of the skids and deckboards. Additional stringers shall be spaced approximately 42 inches apart on center. Intervals between the last two stringers may vary but

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shall not exceed 48 inches. For plate 1/2 inch thick and over, spacing of stringers may be increased but shall be such as to prevent sagging of metal between stringers. The stringers shall be placed so as not to interfere with center side forklift handling.

5.3.1.2.1.6 Deckboards. Deckboards shall be a minimum of 1 X 6 inch (nominal) lumber spaced a maximum of 8 inches apart. For plate 1/2 inch thick and over, spacing of deckboards shall be such as to prevent sagging of metal between deckboards.

5.3.1.2.1.7 Nailing. Nails shall be driven so that the head will not project above the surface of the deckboard. Occasional overdriving of nails will be permitted but no nail shall be overdriven more than 1/2 the thickness of the deckboard. Nails used in construction of pallets shall be Type II, Style 18 of FF-N-105 (pallet nails). Minimum sizes shall be as shown in Table XIII.

TABLE XIII

Minimum Nail Sizes for Pallets (Figs 15-18)

Parts Nailed	Length (inches)
Skids to Stringer (Fig 17, 18)	3 1/2
Deckboards to Skids or Stringers	2 3/8
Rubbing Strips to Skids	3 1/2

5.3.1.2.2 Pallet enclosure construction (Figures 12 and 14).

5.3.1.2.2.1 Sides and ends. One piece sides shall not be less than 1/2 inch thick and one piece ends shall not be less than 3/4 inch thick. When sides and ends exceed 11 1/2 inches in height (Figure 14), they shall be constructed like style 4 box (Figure 3) and shall be not less than 3/4 inch thick.

5.3.1.2.2.1.1 Side and end battens (Cleats). When sides and ends are of Figure 14 construction, minimum 1 X 4 inch battens and cleats shall be used. Side battens shall be placed under girthwise straps. End cleats shall be nailed.

5.3.1.2.2.2 Tops. Tops shall be of minimum 1/2 inch thick lumber, 1/2 inch thick plywood or 1/4 inch thick hardboard.

5.3.1.2.2.2.1 Top battens. Top battens shall be 1 X 4 inch lumber. For lumber tops, battens shall be placed under each girthwise strap. For plywood and hardboard tops, battens shall be used only when the top consists of more than one piece of plywood or hardboard and shall be placed over each joint and underneath each girthwise strap.

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5.3.1.2.2.3 Nailing sides, ends and tops. Sides, tops and bottoms shall not be nailed on pallet enclosures 11 1/2 inches or less in height. When pallet enclosures are of Figure 14 construction, sides, ends and tops shall be nailed in accordance with Tables XI and XII, except that sides, ends and tops for pallets of flat plate, roofing and siding sheets shall not be nailed.

5.3.1.2.2.4 Strapping. Flat steel strapping shall be minimum 3/4 by 0.023 inch for pallets containing 4,000 pounds net weight or less and 1 1/4 by 0.031 inch for pallets containing over 4,000 pounds net weight.

5.3.1.2.2.4.1 Strap spacing. A minimum of two lengthwise straps shall be used for widths less than 60 inches and three lengthwise straps for widths 60 inches and over. Lengthwise straps may be located under skids or under deckboards along side skids. Girthwise straps shall be placed not more than 9 inches from each end. Intermediate straps shall be applied at approximately 42 inch intervals and at plywood and hardboard joints. Girthwise straps shall pass between the deckboards and stringers or skids (Figures 12 and 14).

5.3.1.3 Crates (Figures 20, 21, and 22).

5.3.1.3.1 Style 1 crate (Figure 20). Open crate for net weights not exceeding 700 pounds and interior dimensions not exceeding 48 inches long by 42 inches wide by 48 inches high.

5.3.1.3.1.1 Fabrication. Style 1 crates shall be constructed of 1 X 4 inch lumber framing members, struts and diagonals arranged as shown in Figure 20. Spacing between struts shall be such as to prevent loss of contents. Diagonals may be placed on the inside or outside.

5.3.1.3.1.2 Nailing. Cement coated or chemically etched 7d nails shall be used. Bright nails may be used in lieu of coated or etched nails for groups III and IV woods. Framing members and struts shall be fastened with a minimum of two nails at each end. Diagonals shall be fastened to other members with a minimum of three nails at each end.

5.3.1.3.1.3 Strapping. Flat steel strapping 5/8 inch by 0.020 inch or round steel strapping 0.080 inch (14 gage) shall be used. Equivalent nonmetallic strapping may be used when comparable strength requirements are satisfied. Straps shall be applied perpendicular to edges of the crate tensioned to sink into the wood at edges of crate provided corner protectors are not used.

5.3.1.3.1.3.1 Strap spacing. Two girthwise straps shall be applied, one over each vertical framing member.

5.3.1.3.1.4 Skids. Skids fabricated from nominal 4 X 4 inch lumber or laminated 2 X 4 inch lumber shall be attached perpendicular to the long dimension of all crates with gross weight exceeding 200 pounds. As an alternate, style 1 crates weighing over 200 pounds gross may be strapped to a pallet in accordance with MIL-STD-147.

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5.3.1.3.2 Style 2 crate (Figure 21). Open crate for net weights not exceeding 4,000 pounds and interior dimensions not exceeding 40 feet long X 6 feet wide X 4 feet high.

5.3.1.3.2.1 Fabrication. Style 2 crates shall be constructed with skids, deckboards, rubbing strips, battens, framing members, ends and sling reinforcements, arranged as shown in Figure 21.

5.3.1.3.2.2 Skids. A minimum of two skids shall be used for crates not exceeding 4 feet in width, and a minimum of three skids shall be used for crates 4 feet or greater in width. Skids shall be fabricated from 2 X 4 inch lumber for crates requiring rubbing strips (5.3.1.3.2.3) and 4 X 4 inch lumber for crates not requiring rubbing strips.

5.3.1.3.2.3 Rubbing strips. In addition to skids, rubbing strips shall be added to crates 60 inches or more in length. Rubbing strips shall be fabricated from 4 X 4 inch lumber or laminated 2 X 4 inch lumber and shall be attached to skids in such a manner as to provide 4 way forklift entry. The bottom half of rubbing strip ends shall be beveled at an angle of 45° plus or minus 5°.

5.3.1.3.2.4 Deckboards. Deckboards function as bottom battens for contents of crate. Deckboards shall be 2 X 6 inch lumber, spaced not more than 12 inches from ends of crate and no more than 60 inches apart on centers and shall be secured to skids at each intersection. Deckboards shall extend approximately 1 inch beyond each side of contents.

5.3.1.3.2.5 Battens (Cleats.) Battens shall be of 2 X 6 inch lumber, grooved to retain straps placed around them. Battens shall be spaced identically with deckboards, side battens set on 1 inch extension of deckboards. Side battens shall be approximately 3/8 inch shorter than height of contents of crate. Top battens shall extend approximately 1 inch beyond each side of contents of crates.

5.3.1.3.2.6 Framing members. Framing members shall be 1 X 6 inch lumber and arranged as shown in Figure 21.

5.3.1.3.2.7 Sling reinforcements. Sling reinforcements shall be 2 X 6 inch lumber positioned on each side at each end between the lower longitudinal framing members and contents so as to distribute sling pressure over a greater area. Sling reinforcements shall extend the full distance between two sets of battens.

5.3.1.3.2.8 Ends. Ends shall be constructed of 1 inch lumber with 1 X 4 inch lumber framing or cleats around all edges of end. Ends shall be the same dimensions as width and height of contents of crate plus thickness of deckboards and battens.

5.3.1.3.2.9 Nails. Cement coated or chemically etched 7d nails shall be used. Bright nails may be used in lieu of coated or etched nails for groups III and IV woods. Spacing of nails and number of nails to be used for fastening members together shall be as shown in Table XIV.

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TABLE XIV

Average Nail Spacing (or Number of Nails at Intersection of Members) of Styles 2 and 3 Crates (Figs 21-22)

Parts Nailed	Spacing Inches	Number of Nails at Each Intersection
Deckboards to skids	- - - -	3
Long edges of adjacent framing members	6	- - - -
Ends of framing members to ends	- - - -	3
Framing members to battens or cleats	-- - -	4
Framing members to sling reinforcement	6	- - - -
End framing members to end sheathing	6	2
Rubbing strip to skid	6	- - - -

5.3.1.3.2.10 Strapping. Size of straps shall be as shown in Table XV. Straps shall be positioned girthwise over each set of deckboards and battens prior to application of framing members, ends and sling reinforcements. Straps shall be tensioned to sink into the wood at edges of battens provided corner protectors are not used, but shall not be tensioned to damage contents of crates.

TABLE XV

Strap Sizes for Styles 2 and 3 Crates (Figs 21-22)

Weight of Contents Pounds	Flat Steel Strapping 1/ Inch	Round Steel Strapping Inch
Less than 700	5/8 x 0.020	0.080 (14 gage)
700 to 2,000	3/4 x 0.023	0.105 (12 gage)
2,000 to 4,000	3/4 x 0.031	0.162 (8 gage)

1/ Other equivalent strapping may be used when comparable strength requirements are met.

5.3.1.3.3 Style 3 crate (Figure 22). Open crate for net weights not exceeding 4,000 pounds and interior dimensions not exceeding 40 feet long X 6 feet wide X 4 feet high.

5.3.1.3.3.1 Fabrication. Style 3 crates shall be constructed with skids, cleats, framing members and ends as shown in Figure 22.

5.3.1.3.3.2 Skids. Skids shall be 4 X 4 or laminated 2 X 4 inch lumber placed girthwise, located at each end of crate and spaced a maximum of 60 inches apart.

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5.3.1.3.3.3 Framing members and cleats. Longitudinal framing members, diagonals and cleats shall be 1 X 6 inch lumber and located as shown in Figure 22. Top and bottom panels shall have girthwise cleats spaced a maximum of 30 inches apart. Side panels shall have vertical cleats spaced a maximum of 60 inches apart.

5.3.1.3.3.4 Ends. Ends shall be constructed of 1 X 6 inch lumber. Ends shall be the same dimensions as width and height of contents of crate. Spacing between vertical end boards shall be such as to prevent loss of contents.

5.3.1.3.3.5 Nails. Cement coated or chemically etched 7d nails shall be used. Bright nails may be used in lieu of coated or etched nails for groups III and IV woods. Spacing of nails and number of nails shall be as shown in Table XIV.

5.3.1.3.3.6 Strapping. Size of straps shall be as shown in Table XV. Straps shall be positioned girthwise and centered over side panel framing members. Straps shall be tensioned so as to sink into the wood at corners of crate.

5.3.1.4 Bundles, pallets and coils (Figures 19, 23, 26 and 27). Bundles, pallets and coils shall be prepared as shown in Figures 19, 23, 26 and 27. Pallet size (Figure 26) shall not exceed 40 inches by 48 inches.

5.3.1.4.1 Strapping. Coils shall be tied in a minimum of 3 places with flat strapping after wrapping (Figure 19). Size of strapping for pallets (Figure 26), bundles (Figures 23 and 27) and coils (Figure 19) shall be as shown in Table IX.

5.3.1.5 Reels (Figure 24). Reels shall be constructed as shown in Figure 24. Reels shall be completely enclosed with wood lagging with lags touching each other. Lags shall be 1 1/2 inches thick lumber and shall be nailed to circumference of head of reel extending across edges of both heads. Lags shall be tied with two 0.1620 inch diameter (8 gage) high tensile round straps, 3/4 X 0.025 inch flat steel straps or non-metallic straps of comparable strength.

5.3.1.6 Flanges for supporting coiled foil (Figure 25). Flanges shall be fabricated and positioned as shown in Figure 25. Flanges supporting weights less than 100 pounds shall be 5/8 inch thick. Flanges supporting weights of 100 pounds and over shall be 3/4 inch thick or 5/8 inch thick with 5/8 inch by 2 1/2 inch vertical cleats. Note: Several narrow coils may be suspended on one dowel.

5.3.1.7 Spools. Spools for products other than inert gas welding electrodes shall be standard commercial quality, type and size normally used for aluminum and magnesium products. Spools for inert gas welding electrodes shall be in accordance with Table XVI.

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TABLE XVI

Dimensions for Spools of Inert Gas Welding Electrodes (see Table I)

Spool Part	One Pound Plastic Spool Size 1/ Inches	10 Pound Fiber Spool Size 1/ Inches	12 1/2 Pound Plastic Spool Size 1/ Inches
Width, Overall	1 3/4 $^{+}_{-}$ 1/32	4 $^{+}_{-}$ 1/16	4 $^{+}_{-}$ 1/16
Flange Diameter	4 + 0, - 1/16	12 max ---	12 max ---
Arbor Hole Diameter	0.630 $^{+0.005}_{-0.000}$	2 1/16 $^{+ 1/16}_{- 0}$	2 1/32 $^{+ 1/16}_{- 0}$
Engaging Hole Diameter	None ----	7/16 $^{+ 0}_{- 1/32}$	7/16 $^{+ 0}_{- 1/32}$
Engaging Hole Location	None ----	1 3/4 $^{+}_{-}$ 1/64 (Off Center)	1 3/4 $^{+}_{-}$ 1/64 (Off Center)

1/ Tolerances allowed shall be as indicated.

5.3.1.8 Steel pails. Steel pails shall conform to or be equivalent with the requirements of PPP-P-704.

5.3.1.9 Steel drums. Steel drums shall conform to or be equivalent with the requirements of PPP-D-705 or PPP-D-729.

5.3.1.10 Metal cans. Metal cans shall conform to the requirements of PPP-C-96, Type V, Class 2.

5.3.1.11 Fiber drums. Fiber drums shall conform to or be equivalent with the requirements of PPP-D-723.

5.3.1.12 Spirally wound fiber tubes. Spirally wound fiber tubes shall be fabricated in accordance with PPP-T-495, Type II, Class 1, Style A with wood or metal ends. Crushing strength shall be 250 pounds per inch section in lieu of that specified with a minimum wall thickness of 0.250 inch. The wood ends shall be cut the size of the inside diameter of the fiber tube from nominal 1 inch thick wood stapled to container walls. Staples shall be spaced a maximum of 1 1/2 inches apart.

5.3.1.13 Net weight per container. Unless otherwise specified herein or in the contract or order, the net weight per container shall be the maximum acceptable weight specified in Table I.

5.3.1.13.1 When the contract or order specifies a weight that exceeds the maximum permissible weight specified in Table I, the packing requirements shall be specified in the contract or order.

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5.3.2 Level C (Commercial Packing). When Level C or commercial packing is specified, items may be packed bulk in the same manner employed by the supplier to afford protection against damage during shipment and must, as a minimum, meet carrier rules and regulations. Weight per shipping unit shall be specified by the procuring activity at the time of procurement.

5.4 Marking for shipment. Except as specified herein and in the contract or order, bundles, pallets, containers and loose pieces shall be marked in accordance with MIL-STD-129.

5.4.1 Two units of measure. When the Invitation for Bid or Request for Quotation indicates two units, example: LBS (pounds) and SH (sheets), markings shall include the quantity of both units. This marking does not apply for material shipped loose.

5.4.2 Exterior container identification and contract data. Exterior container identification and contract data markings shall be applied using methods and materials of MIL-STD-129 except Figure 8, 9 and 10 boxes.

5.4.2.1 Figure 8, 9 and 10 boxes. Exterior container identification and contract data markings shall be applied on the left, upper two-thirds of the top of the container and may be printed, stenciled or labeled with not less than 1/8 inch high lettering. In addition, the NSN, quantity and unit of issue shall be applied to at least one side and one end of the container and may be printed or stenciled directly on the container or on tags or labels affixed to the container. Size of markings shall be not less than capital letters of equal height. Markings shall be non-fading and durable.

5.4.3 Precautionary marking. The top of all boxes of flat sheet and pallets of flat plate shall be printed, stenciled, or labeled in not less than 1/8 inch high characters as follows:

CRITICAL CONTENTS KEEP DRY

Do not expose to improper handling, bending, scratching, or gouging. Store in enclosed space. Restore contents of opened or damaged boxes to full protection immediately.

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Custodians:
Army - AV
Navy - AS
Air Force - 84
DSA - DH

Preparing Activity:
Air Force - 84
(Project PACK-0513)

Review Activities:
Army - ME, WC, SM, AV
Navy - SA, YD, SH
Air Force - 69, 70
DSA - IS, CS

User Interest:
Navy - MC, EC
Air Force - 71, 80

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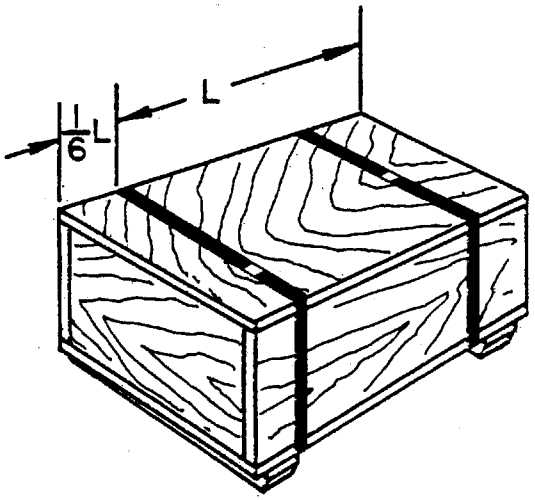


FIGURE 1. Style 1 Nail Wood Box

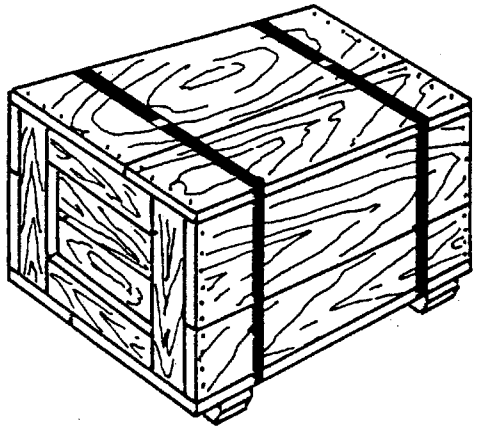


FIGURE 2. Style 2 Nail Wood Box

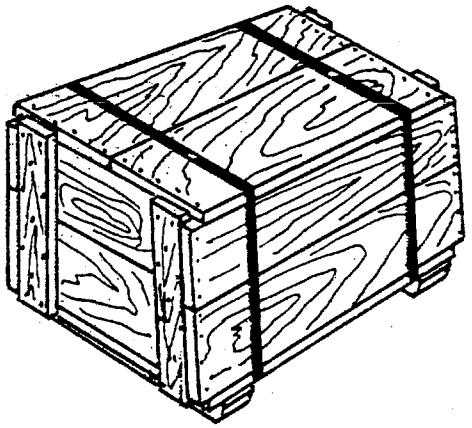


FIGURE 3. Style 4 Nail Wood Box

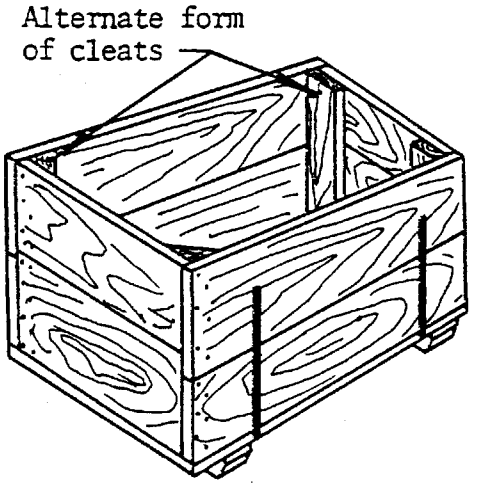


FIGURE 4. Style 5 Nail Wood Box

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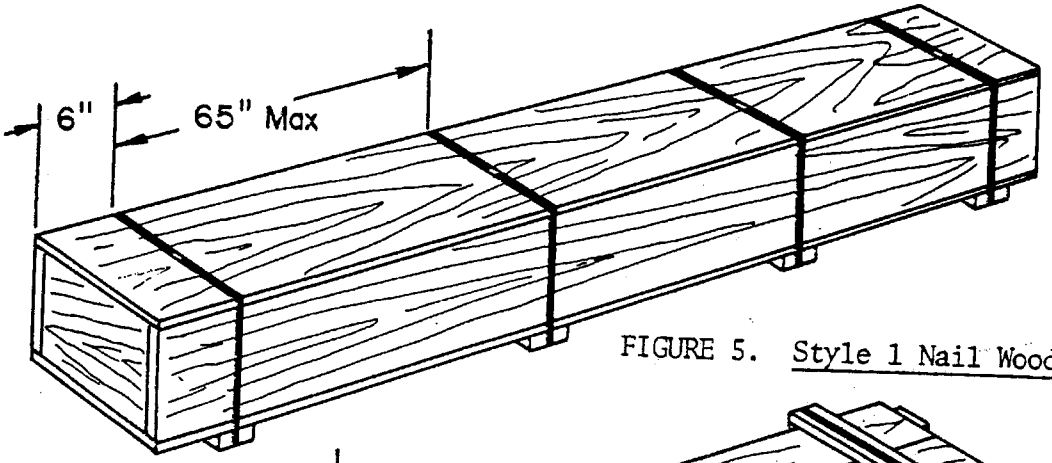


FIGURE 5. Style 1 Nail Wood Box *

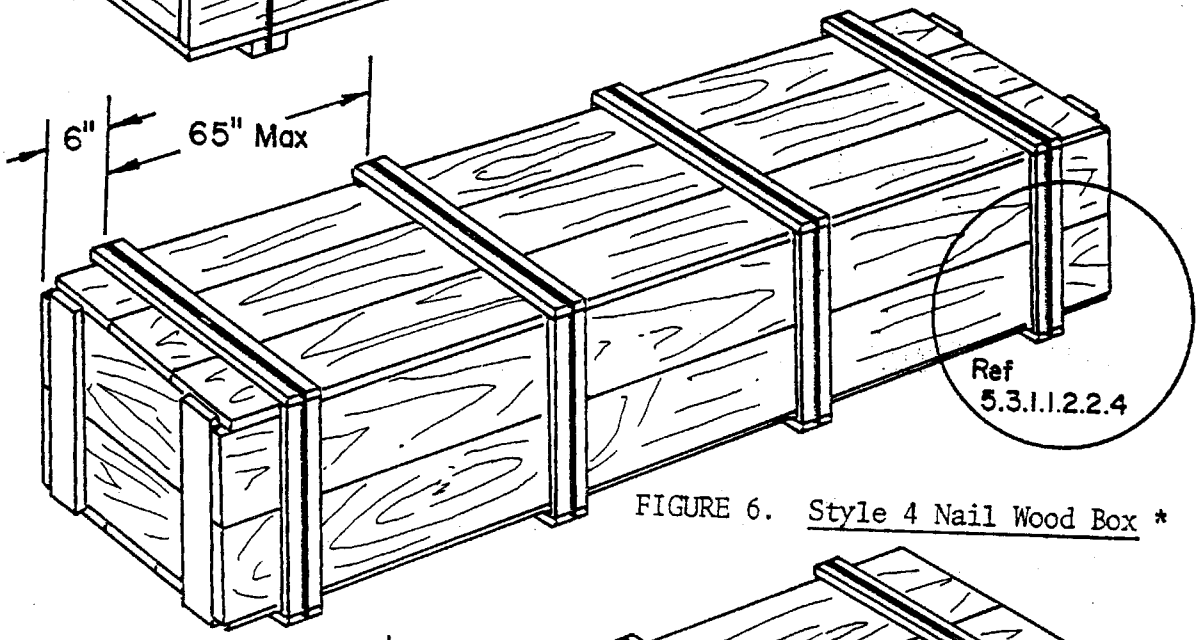


FIGURE 6. Style 4 Nail Wood Box *

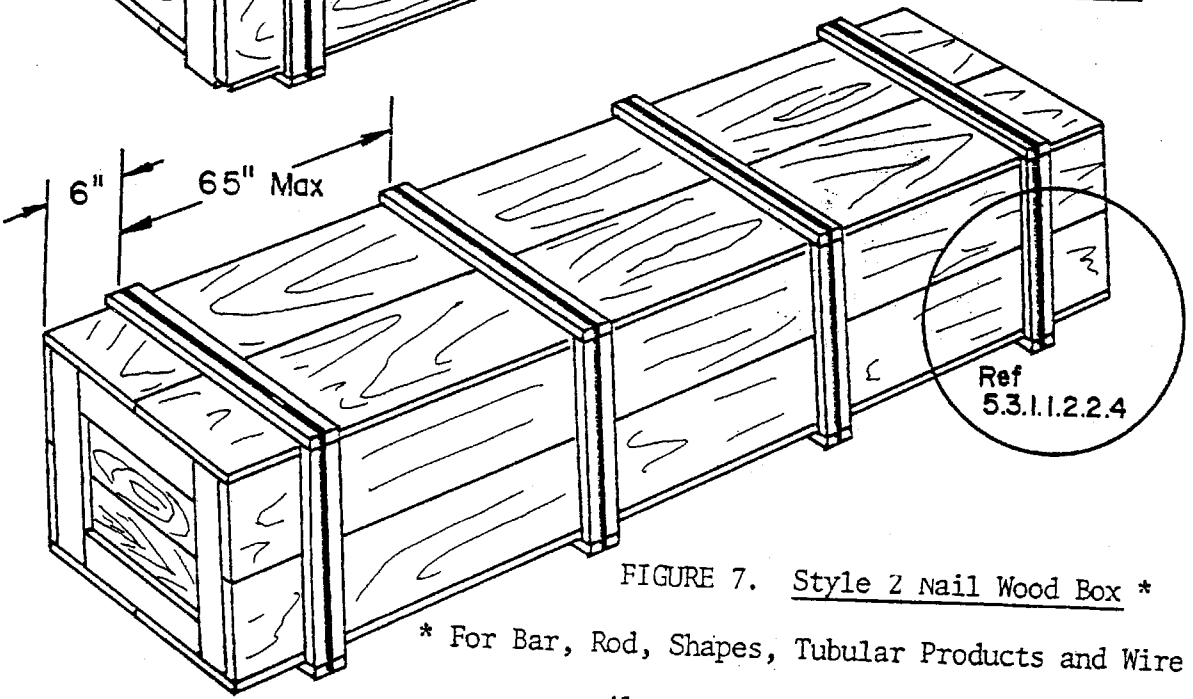


FIGURE 7. Style 2 Nail Wood Box *

* For Bar, Rod, Shapes, Tubular Products and Wire

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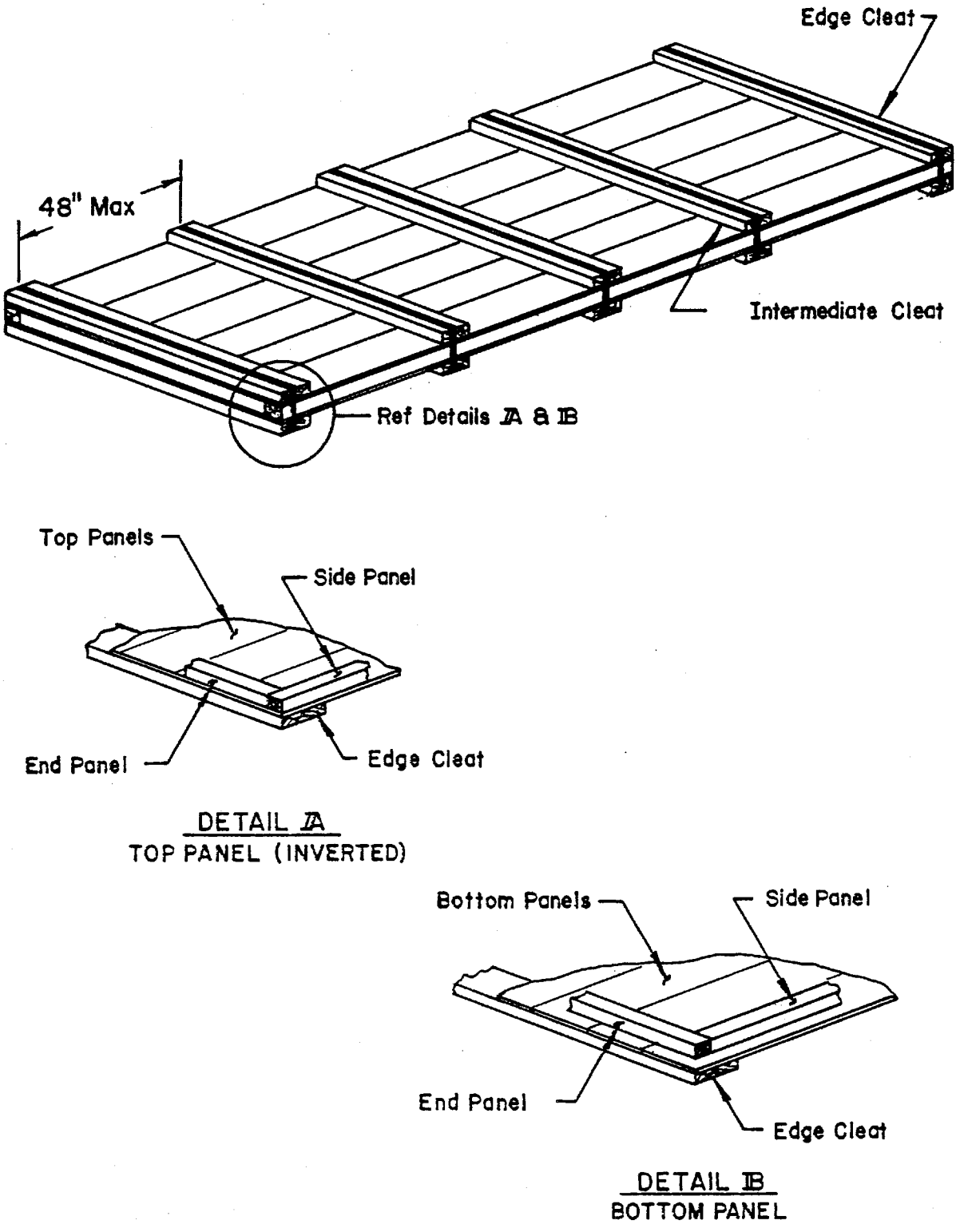
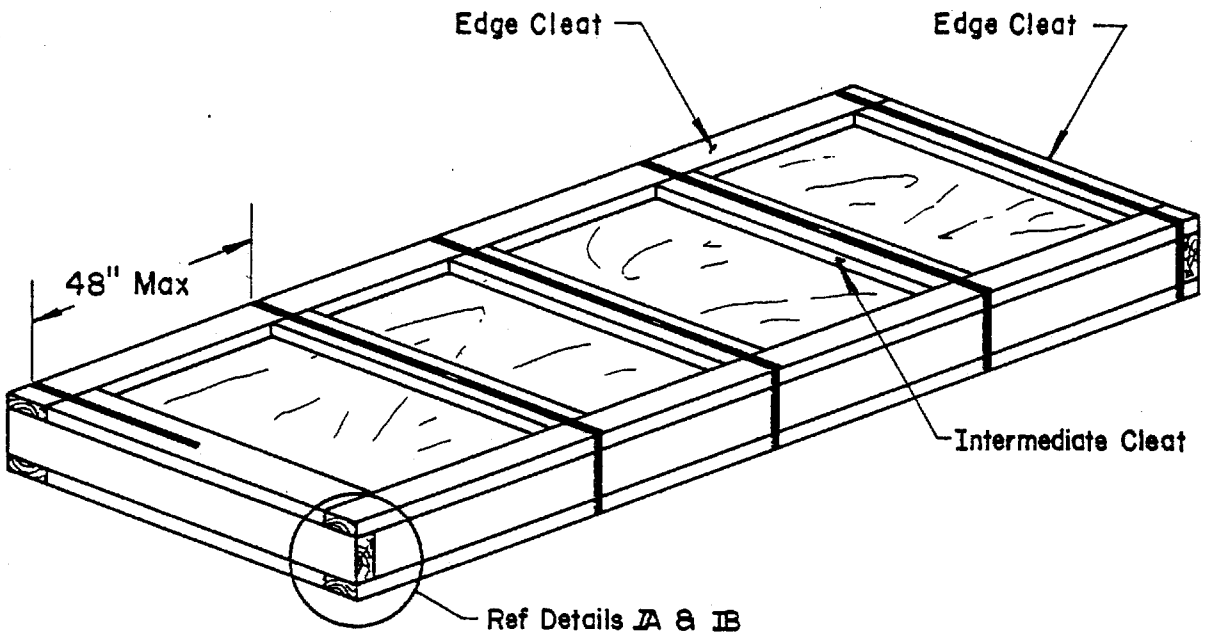
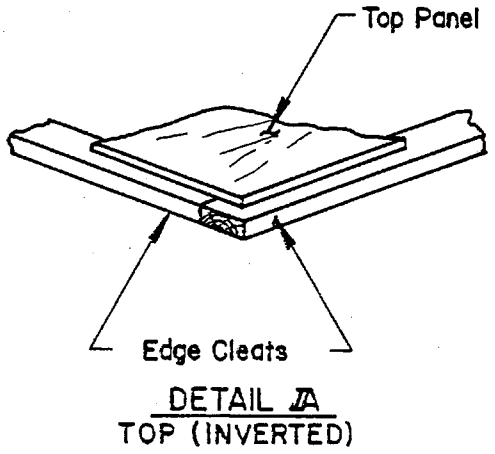


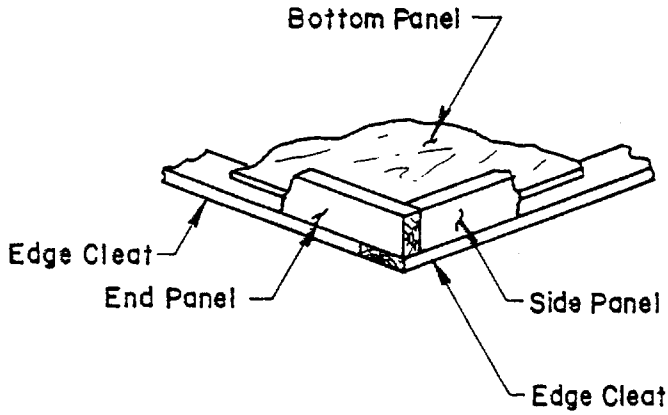
FIGURE 8. Wood Box for Flat Sheet (Telescopic)

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DETAIL ⚠A
TOP (INVERTED)



DETAIL ⚠B
BOTTOM

FIGURE 9. Plywood or Paper Overlaid Veneer Box (Telescopic)

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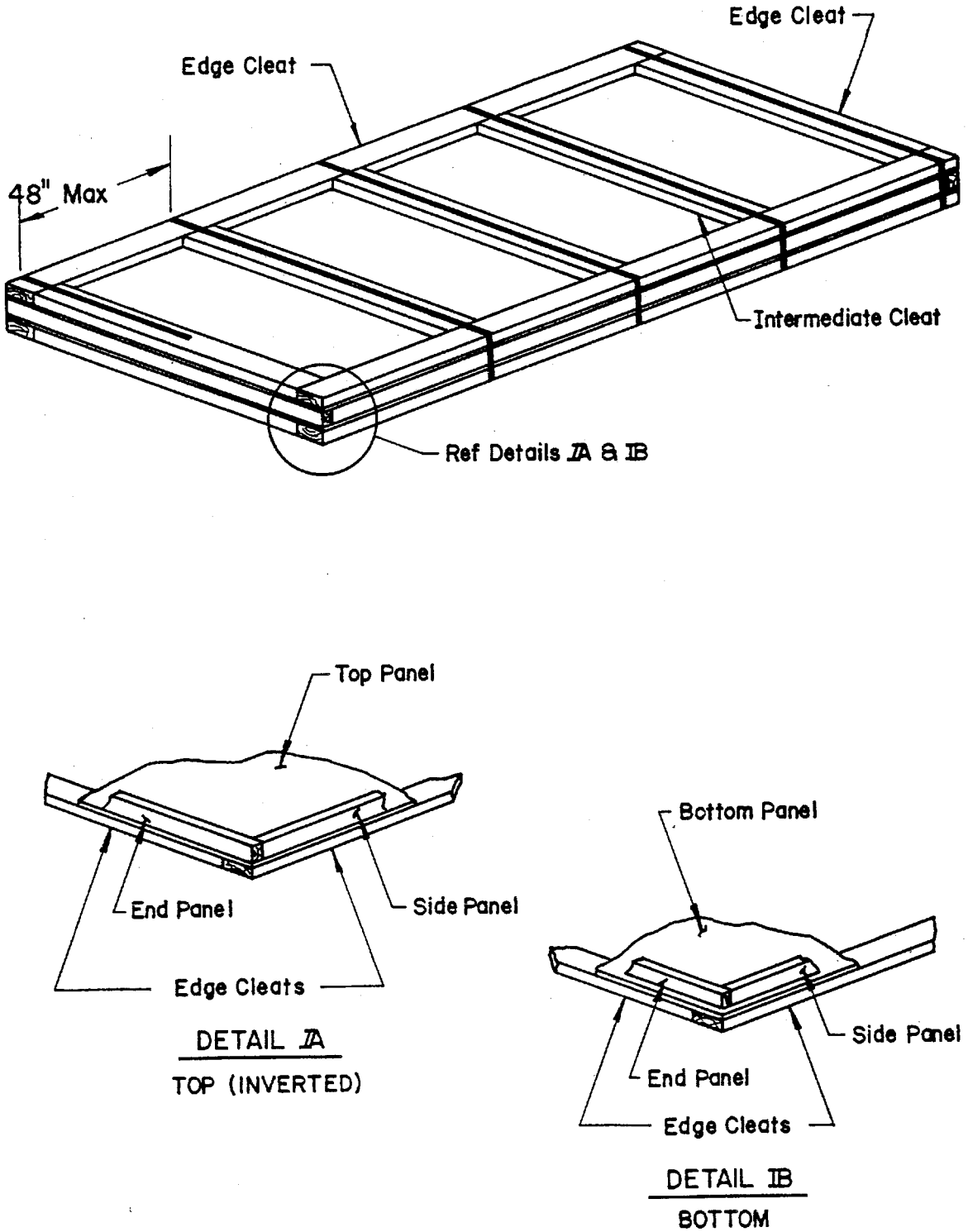
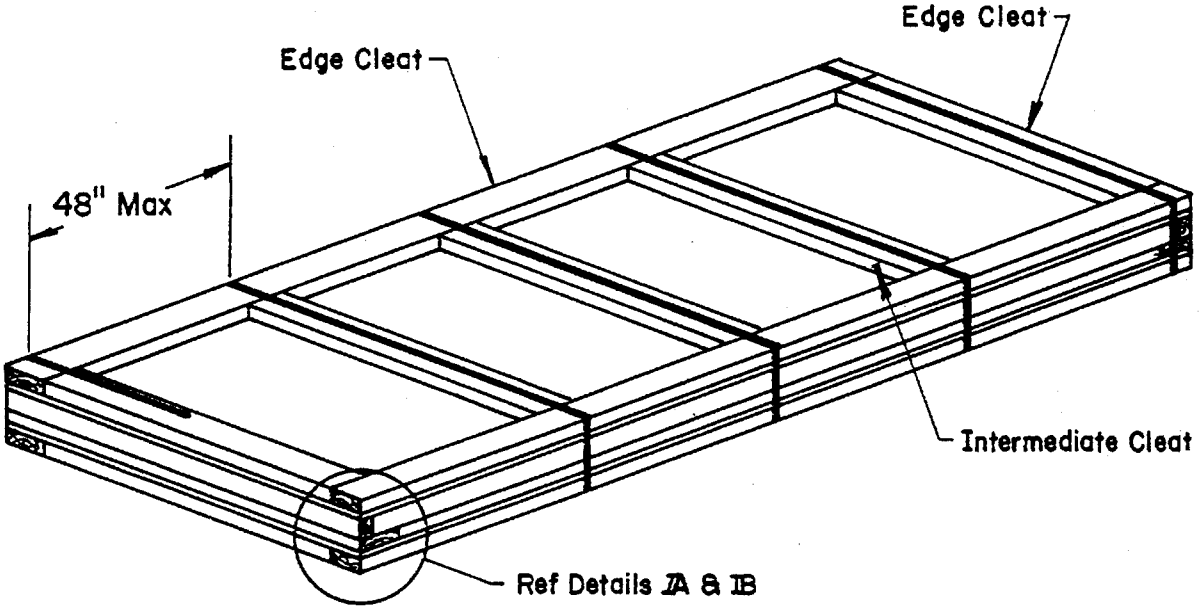
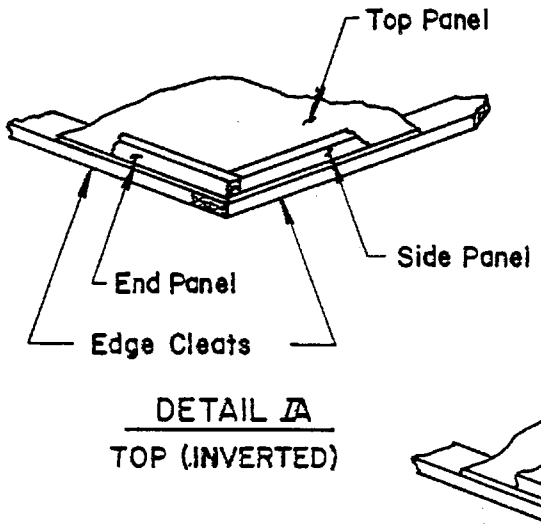


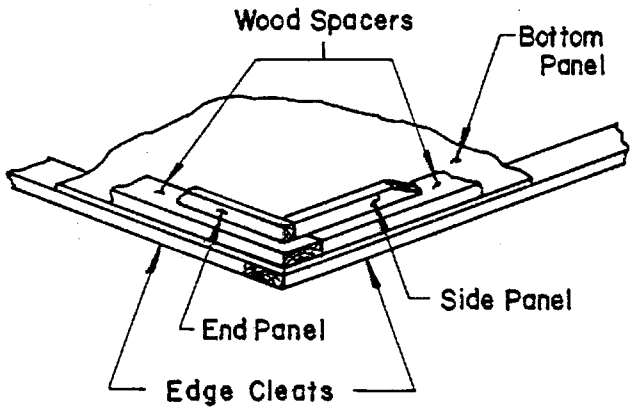
FIGURE 10. Solid Fiberboard Panel Box (Telescopic)

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DETAIL ⒶA
TOP (INVERTED)



DETAIL ⒷB
BOTTOM

FIGURE 11. Solid Fiberboard Panel Box, With Spacer (Telescopic)

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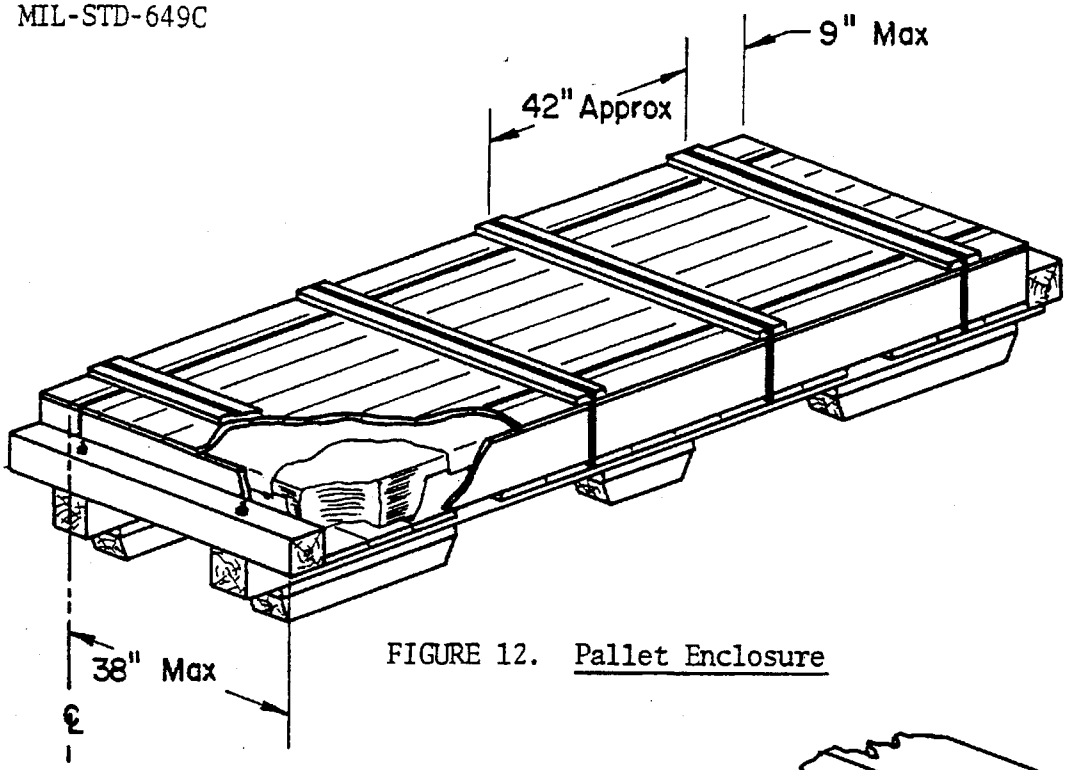


FIGURE 12. Pallet Enclosure

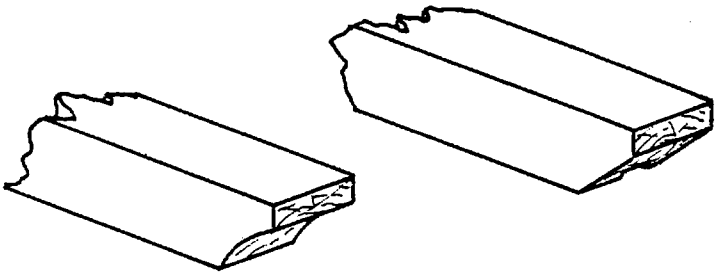


FIGURE 13. Skid Ends Dadoed or Beveled, Bottoms Notched

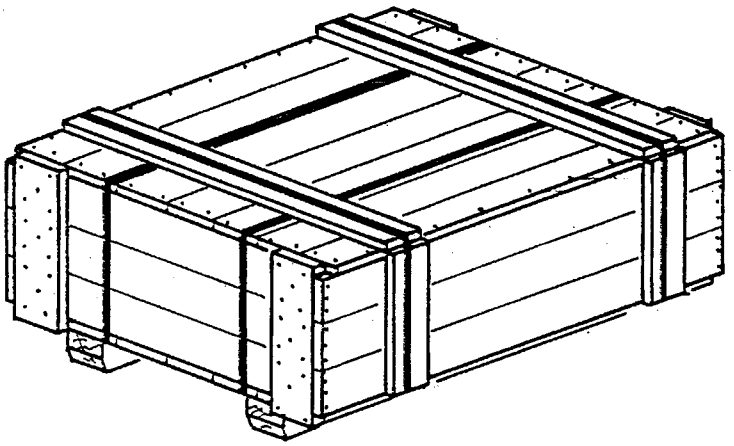


FIGURE 14. Pallet Enclosure

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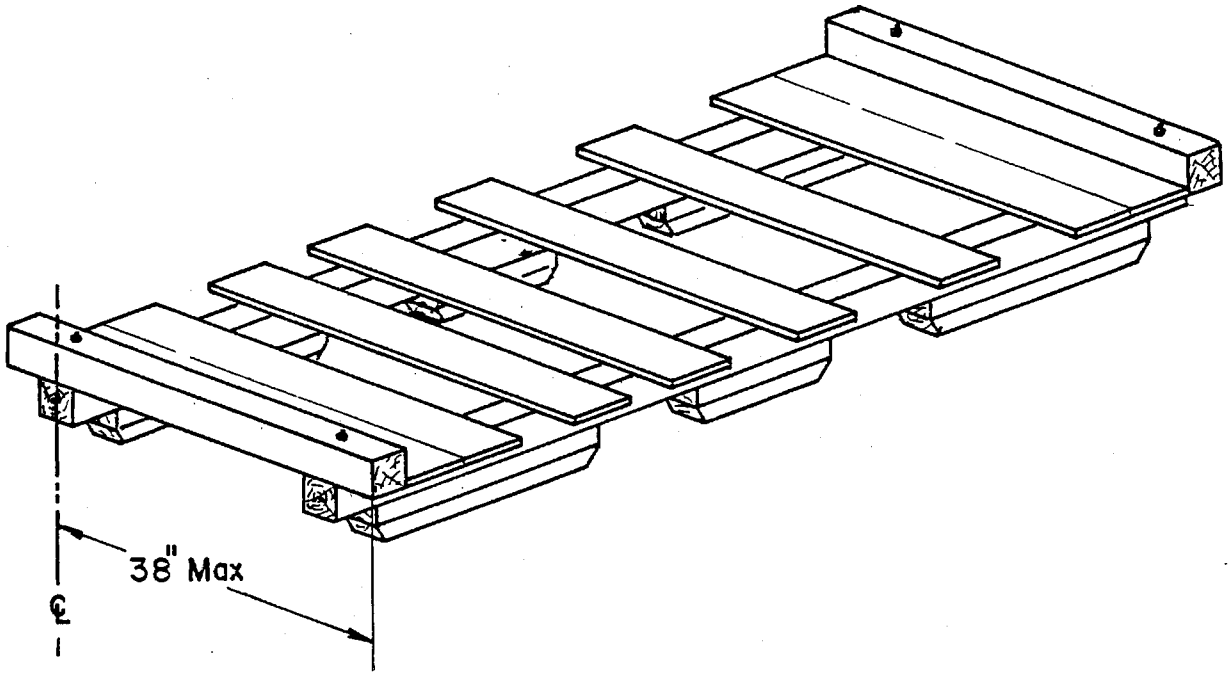


FIGURE 15. Pallet with Lengthwise Skids and Rubbing Strips

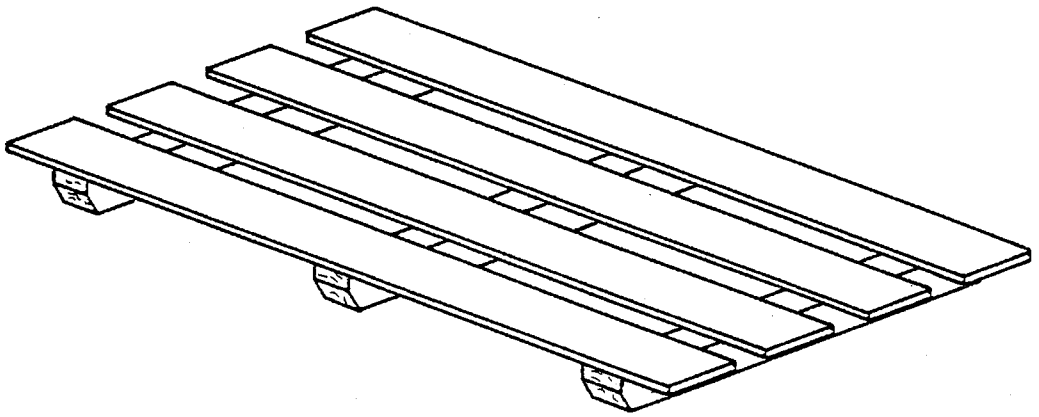


FIGURE 16. Pallet with Crosswise Skids.

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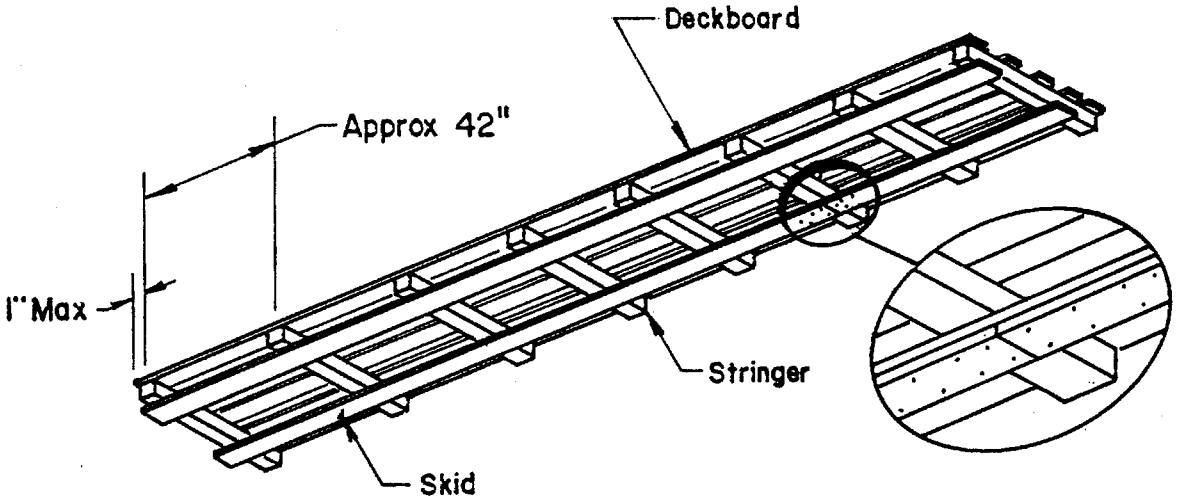


FIGURE 17. Double Faced Pallet with Laminated Skids

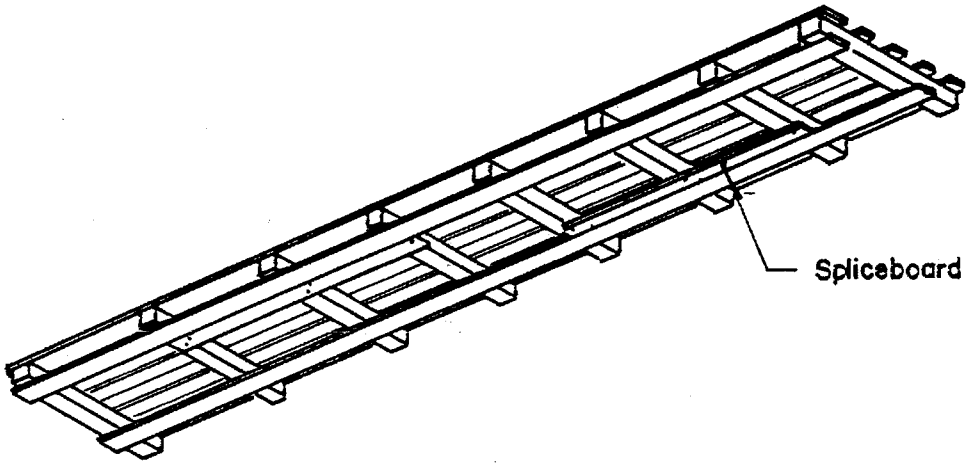


FIGURE 18. Double Faced Pallet with Spliceboards

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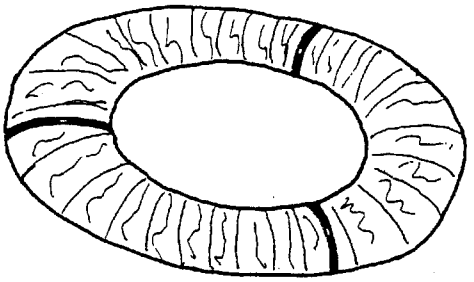


FIGURE 19. Paper Wrapped Coil

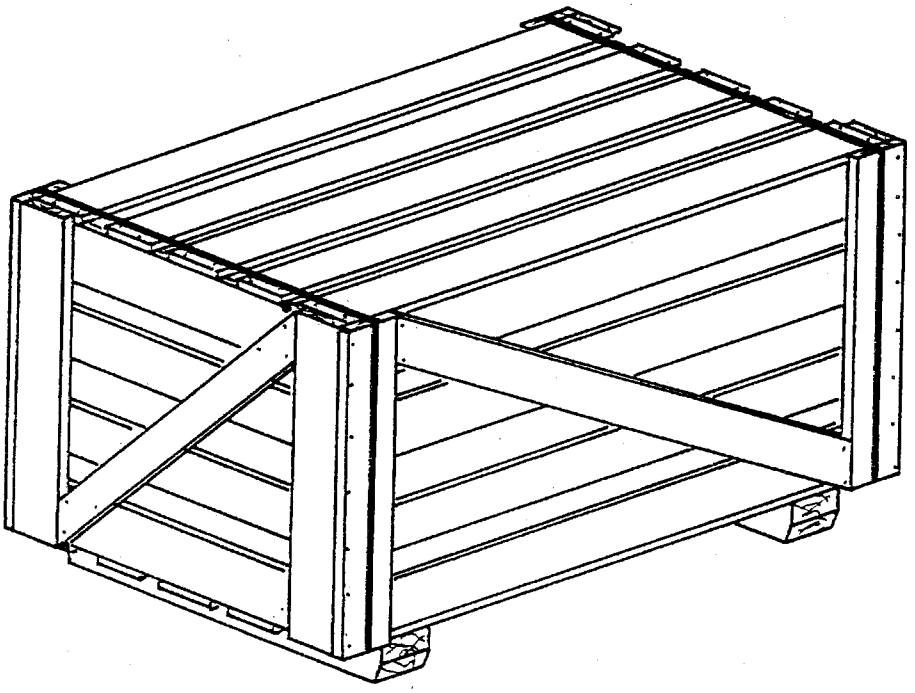


FIGURE 20. Style 1 Crate, open

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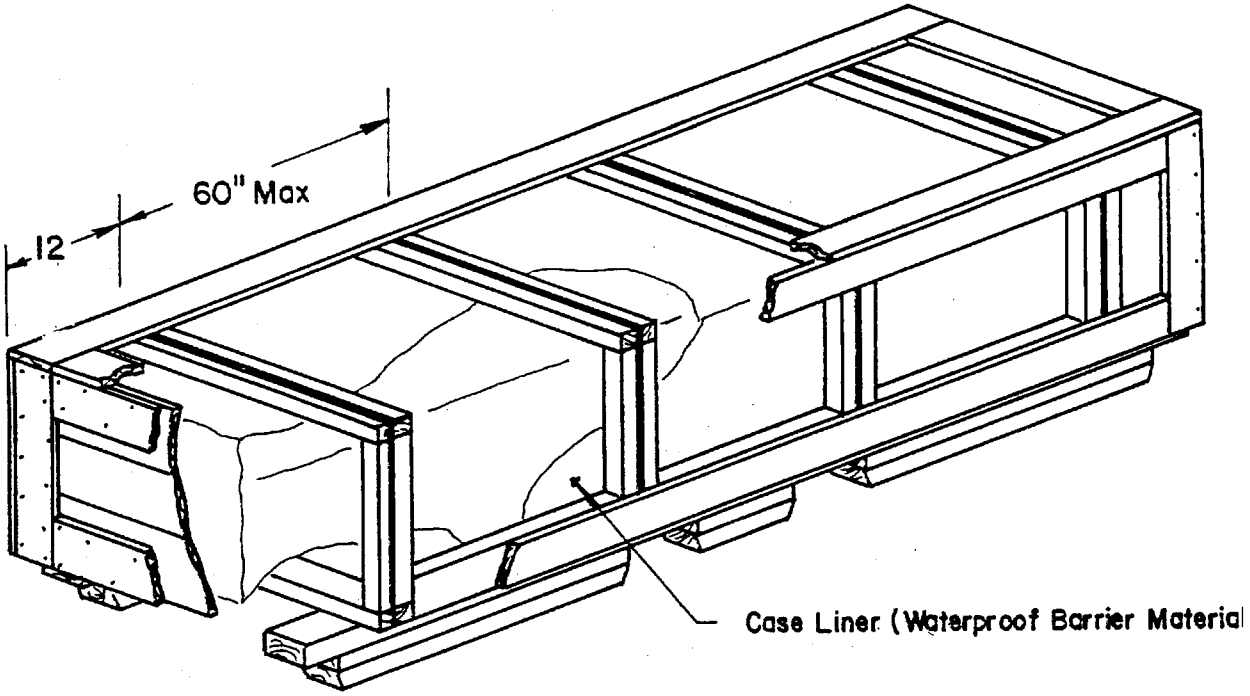


FIGURE 21. Style 2 Crate, open

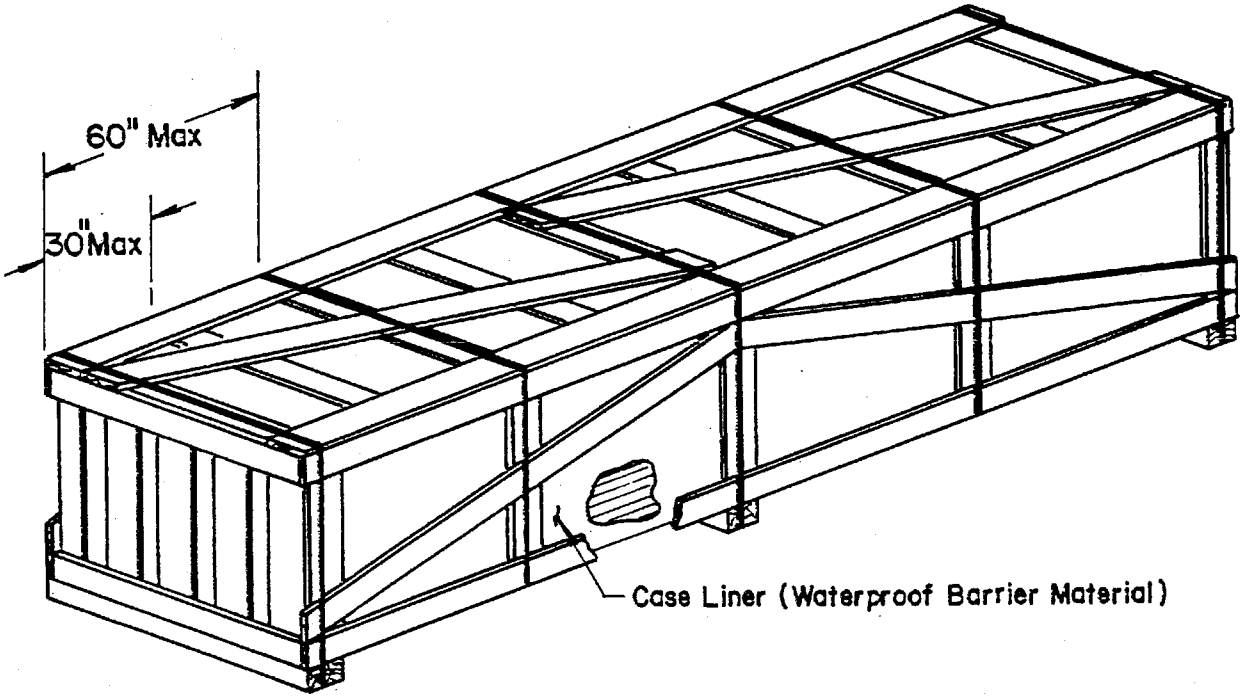


FIGURE 22. Style 3 Crate, open

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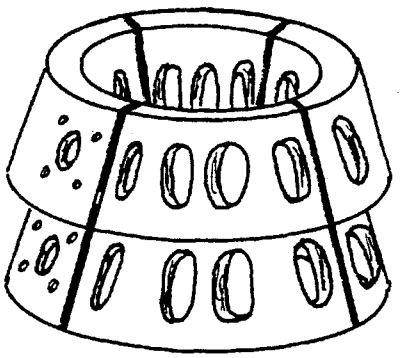


FIGURE 23. Strapped Bundle for Castings and Forgings

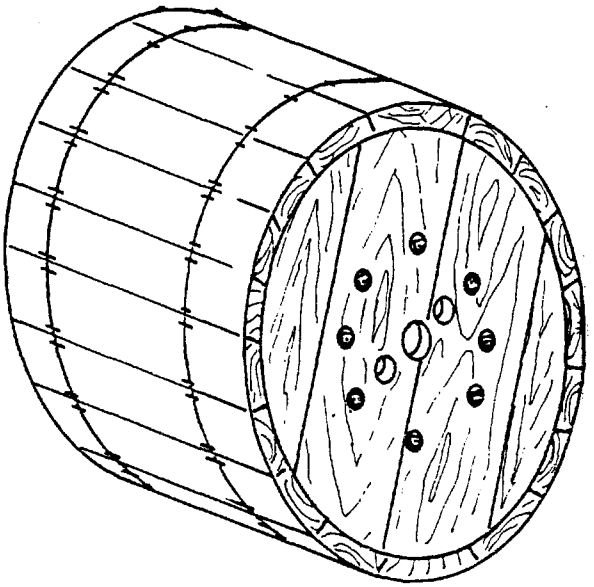


FIGURE 24. Lagged R 1

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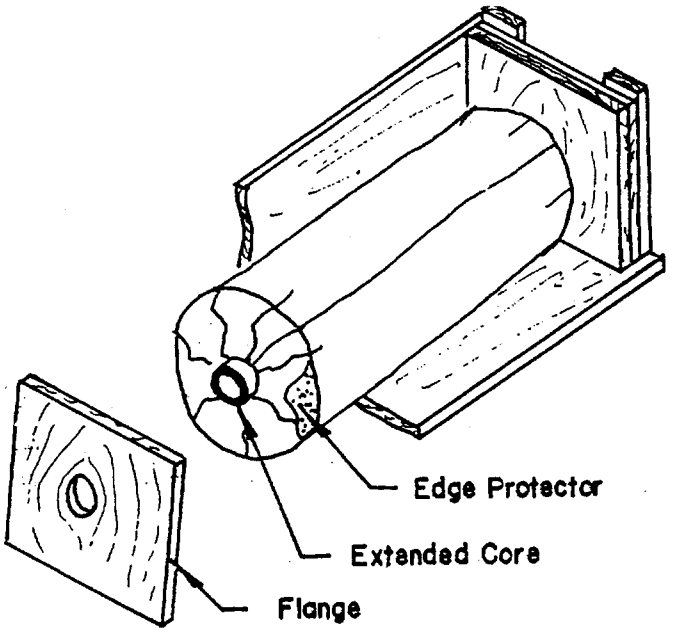


FIGURE 25. Coil Foil Packing

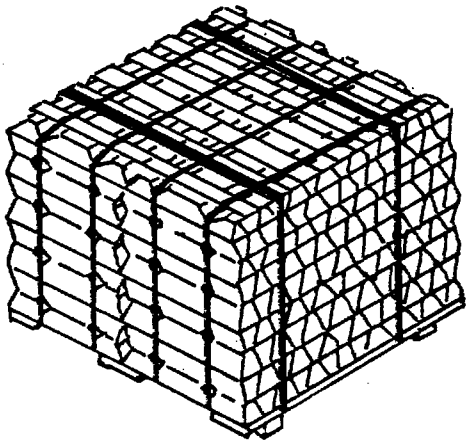


FIGURE 26. Pallet for Ingot

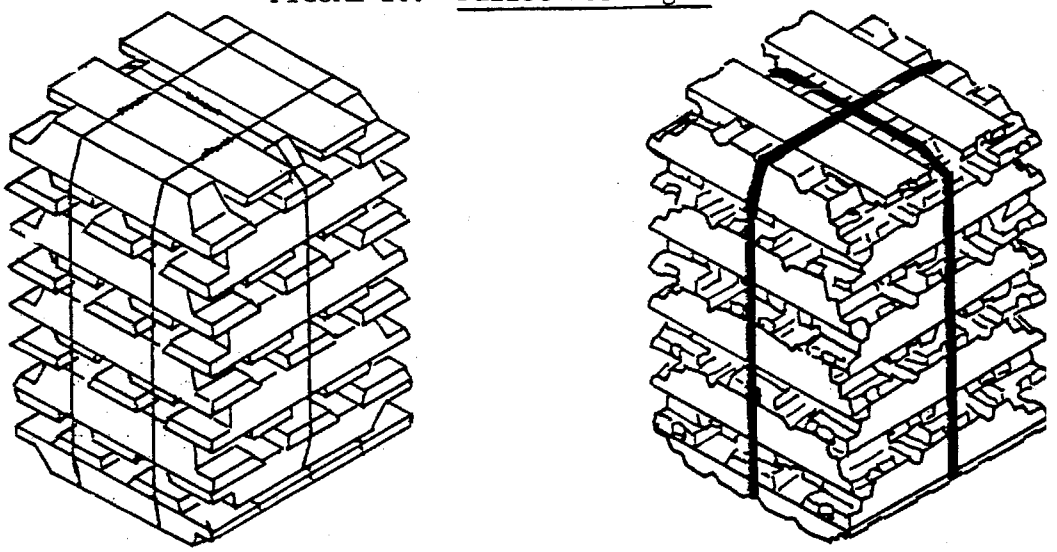


FIGURE 27. Strapped Bundles for Ingot

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